

Master's Thesis

Gradient-Aware Adaptive Rounding for Post-Training Quantization of Deep Neural Networks

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August, 2026

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概要

大規模言語モデル（LLM）の近年の指数関数的なスケール拡大は、莫大な炭素排出量や実装上の課題をもたらすだけでなく、ユーザーがクラウド事業者への依存を強いられるというプライバシー上の問題も顕在化させている。このような状況下、モデル圧縮の重要性が高まっており、その一手法である学習後量子化（Post-Training Quantization、PTQ）はオンデバイスでの効率的な推論を可能にする。適応的丸め手法である AdaRound は、従来の最近接丸めよりも高精度な量子化を実現し、ラベルなしキャリブレーションデータを用いて元のパラメータと量子化後のパラメータの間のタスク損失を最小化する。本論文では以下の二点を明らかにする。第一に、AdaRound はキャリブレーションデータに由来する情報を量子化によるパラメータの変動量に部分的に保持しており、この行列の零空間に着目した新たな逆転攻撃を提案することで、キャリブレーションデータの平均値を粗粒度な復元が可能であることを実証する。第二に、AdaRound の理論的基盤では事前学習済みかつ収束したモデルにおいて損失関数の勾配がゼロであることが仮定されているが、この前提自体が近年の研究によって否定されている点である。この理論的乖離に対処すべく、局所的な勾配信号を AdaRound の最適化プロセスに組み込むことを試みたものの、実験的な評価では有意な改善は認められなかった。以上より本論文は、PTQ 分野において従来等閑視されがちであったプライバシー保護の観点に光を当てるとともに、最新の知見に整合する勾配認識型の AdaRound 改良手法を提案するものである。

Abstract

The recent exponential scaling of Large Language Models (LLMs) has resulted in massive carbon emissions and significant challenges in deployment, forcing users to rely on third-party cloud providers, thereby raising privacy concerns. This highlights the necessity of model compression, and post-training quantization (PTQ) is one method that enables efficient on-device models. AdaRound, an adaptive rounding method that outperforms round-to-nearest rounding during quantization, minimizes the task loss between the original weights and the quantized weights on unlabelled calibration data. We find that: 1) AdaRound leaks some information about its calibration data into the weight perturbation matrix, and we demonstrate this by proposing a novel inversion attack that targets the null space of this matrix to recover coarse representations of the original data. 2) The theoretical foundation of AdaRound asserts that the gradient of the loss function is zero for pretrained converged models, an assumption which is refuted by recent literature. To address this discrepancy, we attempted to reconcile this by incorporating local gradient signals into the AdaRound optimization process, but obtained neutral results on empirical evaluations. Therefore, this thesis highlights a frequently overlooked dimension in PTQ (privacy), and proposes a gradient-aware modification of AdaRound to better align the method with modern literature.

Contents

Chapter 1	Introduction	1
1.1	Background	1
1.2	Purpose	3
1.3	Contributions	4
Chapter 2	Preliminary	6
2.1	Compressibility of Deep Neural Networks	6
2.2	Small Language Models	9
2.3	Baseline Round-to-Nearest Quantization	11
2.4	Quantization is Weight Perturbation	12
2.5	Effects of Perturbation of Model Weights	13
Chapter 3	Literature Review	18
3.1	AdaRound	18
3.2	Edge of Stability and Central Flows	21
Chapter 4	AdaRound Inversion Algorithm	23
4.1	Formulation of Leakage Channel	23
4.2	Method	23
4.3	Evaluation	24
4.4	Ablation Study	26
4.5	Discussion	27
Chapter 5	Gradient-Aware Adaptive Rounding	29
5.1	Method	29
5.2	Evaluation	30
5.3	Ablation Study	31
5.4	Discussion	32
Chapter 6	Conclusion	33
	Acknowledgements	34
	Bibliography	35

Chapter 1

Introduction

1.1 Background

Recently, the exponential scaling of LLM sizes (as shown in Figure 1.1) has inadvertently resulted in models that are extremely challenging to deploy, thus causing users to rely on third-party providers for LLM usage. This resulted in the effects including but not limited to the following.

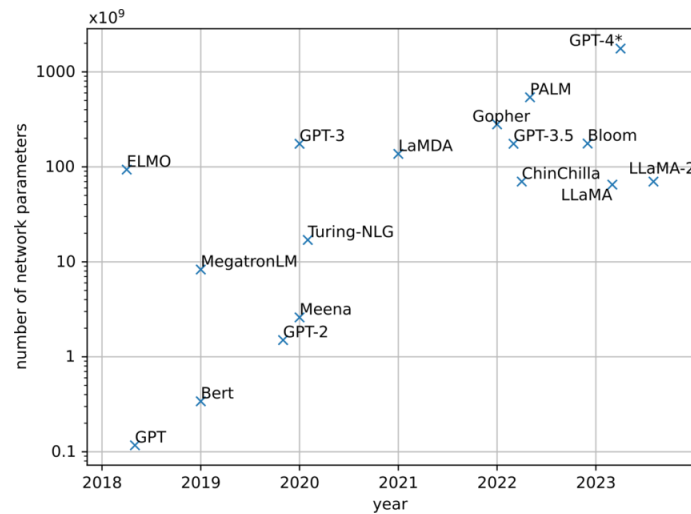


Fig. 1.1: Observed exponential scaling of LLM sizes over time [1] (Note: The GPT-4 figure is an estimate)

Environmental Effects. CO_2 -equivalent emissions (CO_2e) seems to also grow over time. GPT-3’s training had an estimated CO_2e of ~ 552.1 tons [2], GPT-4’s training had an estimated CO_2e of ~ 6912 tons [3], and Llama 3.3 [4]’s training had an estimated CO_2e of $\sim 11,390$ tons. Even worse, it is shown that an LLM incurs “significantly higher carbon emissions” during inference than training [5].

Cost. As LLMs scale exponentially, inference costs also become expensive. For example, OpenAI o1 [6] in “pro mode” costs \$150 per 1 million tokens input, and \$600 per 1 million tokens output. We see evidence of heavy AI cost burden in companies: Uber’s yearly AI

budget was exhausted in 4 months, while Microsoft began cancelling Claude Code licenses just 6 months after acquiring them [7].

Reliability. Using third-party providers can cause a single point of failure, and an outage can have severe effects on mission-critical tasks. For example, it is observed that whenever ChatGPT has an outage, stock trading volume goes down [8].

To further understand why simple scaling seems to improve model accuracy with little drawback, let us examine the clash between the classical Bias-Variance Tradeoff and the modern Scaling Laws.

Classical Perspective

Consider a classifier over $k \in \{1, 2, \dots, K\}$, trained with Mean Squared Error loss. Its test error is often attributed to the following three sources [9].

$$\mathcal{E}_{\text{total}} = \mathcal{E}_{\text{noise}} + \mathcal{E}_{\text{bias}} + \mathcal{E}_{\text{variance}} \quad (1.1)$$

The first term is an intrinsic error term representing irreducible noise, the second term measure the systematic error the model makes, and the third term captures the variance in the model's predictions. As the capacity of a model increases, the bias decreases while the variance increases, leading to a parabolic test error curve [10]. It is said that:

The price to pay for achieving low bias is high variance. — Geman et. al., 1992 [9]

This means that if the model's capacity is too small, it may underfit the training data and perform poorly on unseen data. If the model's capacity is too large, it may overfit to spurious patterns in the training data, resulting in low accuracy on unseen data anyways [11]. Textbooks even assert that “a model with zero training error is overfit to the training data and will typically generalize poorly” [12, pg. 221]. It was therefore common to carefully select the complexity of models to achieve the optimal balance between bias and variance.

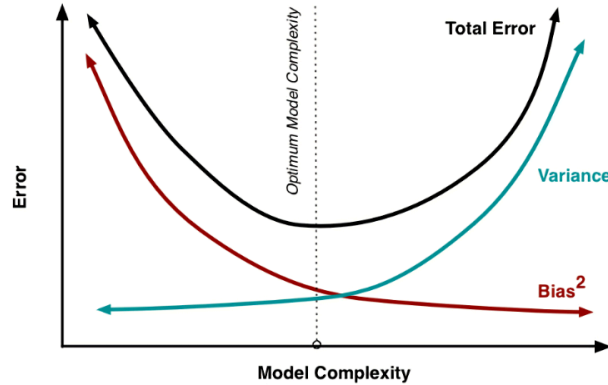


Fig. 1.2: Bias-Variance Tradeoff [10]

Modern Perspective

Modern machine learning, however, regularly uses models with low or zero training error, and those models often give very accurate predictions on new data [11]. It was then proposed to reconcile this phenomenon and the classical bias-variance tradeoff using a “double descent” curve, as shown in Figure 1.3.

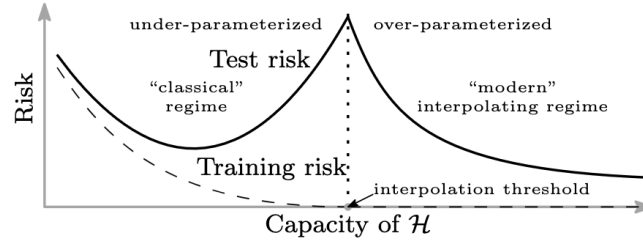
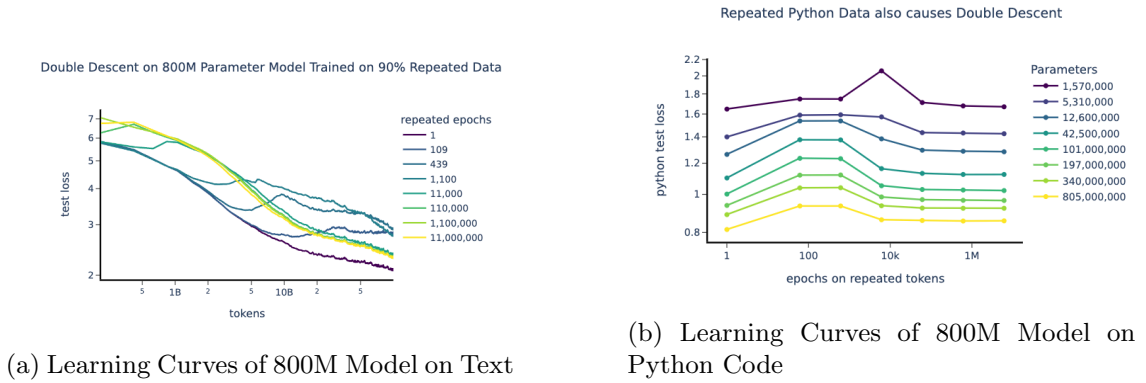


Fig. 1.3: Double Descent Curve [11]

When originally proposed in 2019, this phenomenon was empirically validated on neural networks and random forests on the MNIST dataset [11]. But importantly, it was later empirically validated on a transformer-based large language model with up to 800M parameters [13]: we can clearly observe a double-descent curve in Figure 1.4. The authors also propose the theory that the peak in test loss in the middle of the learning curve is the model memorizing the training data, while the continued descent afterwards is the model actually learning from the training data.



(a) Learning Curves of 800M Model on Text

(b) Learning Curves of 800M Model on Python Code

Fig. 1.4: Double Descent Phenomenon on Transformer-based Language Models [13]

1.2 Purpose

Given the recent exponential scaling of LLMs, we observe an important need for efficient language models that can be run on-device, which provides benefits such as the following.

Privacy. Data stays on the device and is not transmitted to a remote server. For regulated areas like medical, legal, or financial data, this can be a requirement.

Robustness. Even without Internet connection, on-device models can still function. This is suitable for deployment scenarios with little or no internet access (e.g. robotics or drones in remote environments), and mission-critical cases that need to function even during an internet outage.

Reduced Latency. The queueing rules can be set depending on the deployment scenario, instead of relying on third-party provider scheduling, which allows the prioritization of tasks that require low latency.

Reduced Cost and Emissions. Using a third-party provider often requires paying per token, in addition to any network costs. Efficient on-device models can reduce the effective per-token cost and can eliminate network costs. Less emissions can also be expected as a benefit. For example, Dell claims a 87% cost saving over two years by running agents entirely on-premise [14], while Shopify deploys finetuned Qwen and Llama models in their systems [15].

1.2.1 Survey of Compression Methods

The following methods are typically used for the compression of deep neural networks.

Weight Pruning. Model size can be shrunk, and inference can be made compute-efficient, by setting some weights to zeroes, resulting in sparse weight matrices [16].

Knowledge Distillation. It is shown that by making a smaller model learn to match output logits of a larger model (or an ensemble of models), parts of the knowledge in the largest model can be transferred to the smaller model [17].

Weight Palettization. This method clusters weights (e.g. via k -means), then stores every weight as the centroid of the cluster they belong to [18].

Weight Quantization. We focus on this method for our thesis. This method stores weights in a data type of less bit width (e.g. `int8`), compared to the original data type (e.g. `float32`) [19].

1.3 Contributions

Our contributions in this work are twofold: First, we demonstrate a novel inversion method against AdaRound to recover coarse outlines that represent its calibration data, motivating the need to consider privacy in data-dependent quantization method. Second, we propose a gradient-aware adaptive rounding method based on AdaRound, and empirically validate the proposed method on two language models.

The rest of this thesis is organized as follows:

- In Chapter 2, we provide background experiments that show the potential of compression of deep neural networks. We pretrain small language models and use them as a base for exploratory experimentation.

- In Chapter 3, we review AdaRound [20], an existing method for superior weight rounding during quantization, and discover two limitations of the work: a privacy leakage, and a zero-gradient assertion that is refuted by Central Flows [21], a more recent work.
- In Chapter 4, we realize an inversion attack against AdaRound, which provide empirical evidence that some information of its calibration data is leaked into the weight perturbation matrix.
- In Chapter 5, we attempt to reconcile AdaRound’s assertion with Central Flows by incorporating gradient signal into AdaRound.
- In Chapter 6, we conclude our findings and list possible future research directions.

Chapter 2

Preliminary

2.1 Compressibility of Deep Neural Networks

In this section, we describe a simple way to quantify the compressibility of models. The source coding theorem [22] states that:

- n i.i.d. (independent and identically distributed) random variables, each with entropy $H(X)$, can be compressed to $nH(X)$ bits on average, with negligible information loss, as $n \rightarrow \infty$
- Conversely, if they are compressed to fewer bits on average, loss of information must occur.

In this case, $H(X)$, the entropy of a random variable, is the expected amount of information needed to describe it, and can be computed by the following.

$$H(X) := - \sum_{x \in X} p(x) \log_2 p(x) \quad (2.1)$$

Note that the log base determines the units: $\log_2$ yields bits, $\log_{10}$ yields dits, and $\ln$ yields nats.

Consider a model with weights W , each stored in b bits, and there are n weights in total. The total number of bits occupied by the model is therefore $n \cdot b$. This is the capacity of the model: it is the maximum amount of information the weights could carry if every possible bit pattern were equally likely and the weights were statistically independent.

In practice, however, models rarely use their full capacity. We can measure this by treating its weights as independent draws from a categorical distribution over these possible values.

$$\mathcal{X} = \{0, 1, \dots, 2^b - 1\} \quad (2.2)$$

We can use Maximum Likelihood Estimation (MLE) to estimate the distribution as

follows.

$$\hat{p}(x) = \frac{1}{n} \sum_{k=1}^n \mathbf{1}(W_k = x), \text{ where } \mathbf{1}(p) = \begin{cases} 1 & \text{if } p \\ 0 & \text{if } \neg p \end{cases} \quad (2.3)$$

And we can estimate the entropy per weight as follows.

$$\hat{H}(W) := - \sum_{x \in \mathcal{X}} \hat{p}(x) \log_2 \hat{p}(x) \quad (2.4)$$

Note that if a certain value does not appear at all in the model weights, there will be an undefined term $0 \ln 0$ in the above calculation. We use a convention in modern information theory to mitigate this, by using an extension of $x \ln x$ in the form of the following [23, eq. 1.1.2].

$$f(x) = \begin{cases} 0, & x = 0 \\ x \ln x, & x > 0 \end{cases} \quad (2.5)$$

This allows us to treat $0 \ln 0$ as 0. We can see that the total information stored in the weights would be approximately $n \cdot \hat{H}(W)$ bits. But the maximum possible total entropy of the model is $n \cdot b$ bits (achieved by the uniform distribution), it therefore follows that the capacity utilization by a model is the following.

$$U(W) := \frac{\hat{H}(W)}{b} \quad (2.6)$$

Note that $U \in [0, 1]$, and $U = 1$ implies the model is nearly incompressible losslessly, while $U < 1$ implies the model can be losslessly compressed to $1 - U$ times its current size. For example, if $U = 0.5$, the model can be compressed to half its size with negligible information loss.

We now have a method to quantify the compressibility of models. We use it to evaluate several language models, and report the result in Table 2.1.

Table 2.1: Entropy Estimate of Language Models (bfloat16)

Model	$\hat{H}(W)$ in bits	$U(W)$
Qwen 3 4B [24]	10.58	$10.58/16 \approx 66\%$
Qwen 3.5 4B [25]	10.56	$10.56/16 \approx 66\%$
Gemma 4 12B Instruct [26]	10.69	$10.69/16 \approx 67\%$
GLM 4.7 Flash [27] (30B)	10.49	$10.49/16 \approx 66\%$

Interestingly, most of these models use around $\frac{2}{3}$ of their capacity. Let us investigate further on Qwen 3 4B. The model is provided in three data types (bfloat16, fp8, int4), and we use the same method to evaluate their compressibility. We report the results in Table 2.2.

Table 2.2: Entropy Estimate of Qwen 3 4B Weights

Data Type	$\hat{H}(W)$ in bits	$U(W)$
bfloat16	10.58	$10.58/16 \approx 66\%$
fp8	6.63	$6.63/8 \approx 82.9\%$
int4	3.67	$3.67/4 \approx 91.8\%$

We observe that the bfloat16 model, which is ~ 8 GB on disk, only uses $\sim \frac{2}{3}$ of its capacity. This implies that the model can be compressed to ~ 5.33 GB with negligible information loss. This degree of compressibility motivates us to perform an investigation of the distribution of the weights by plotting their possible values against the probabilities of each value, which we report in Figures 2.1, 2.2, and 2.3.

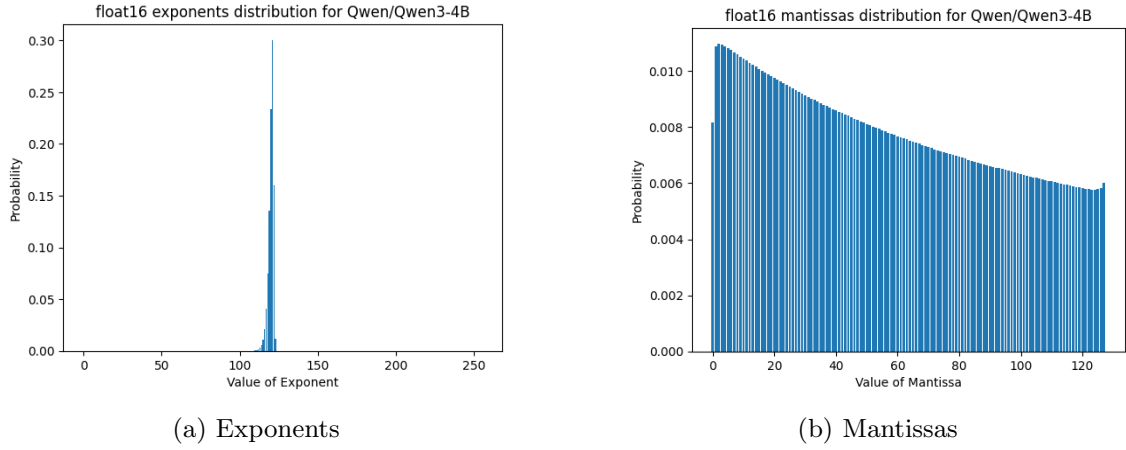


Fig. 2.1: Distribution of Qwen3-4B Weights (bfloat16)

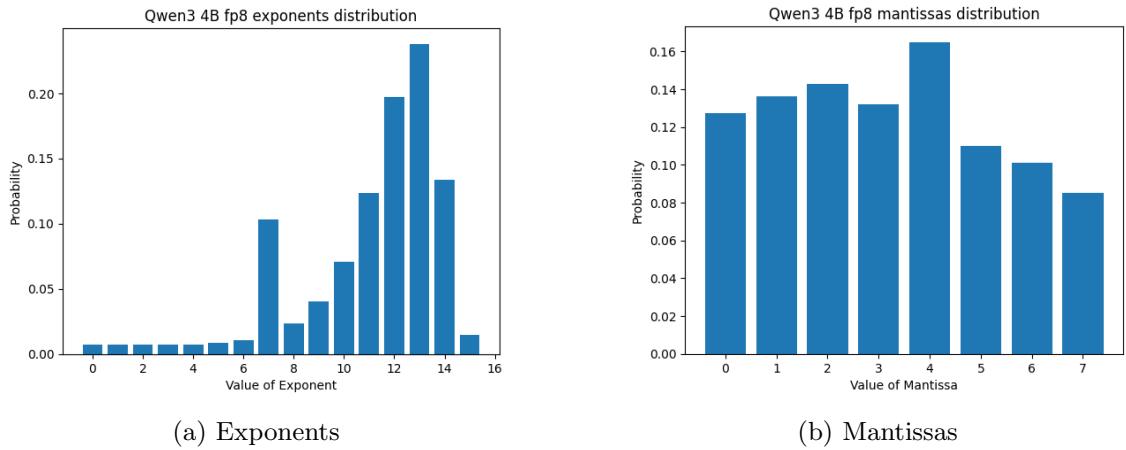


Fig. 2.2: Distribution of Qwen3-4B Weights (fp8)

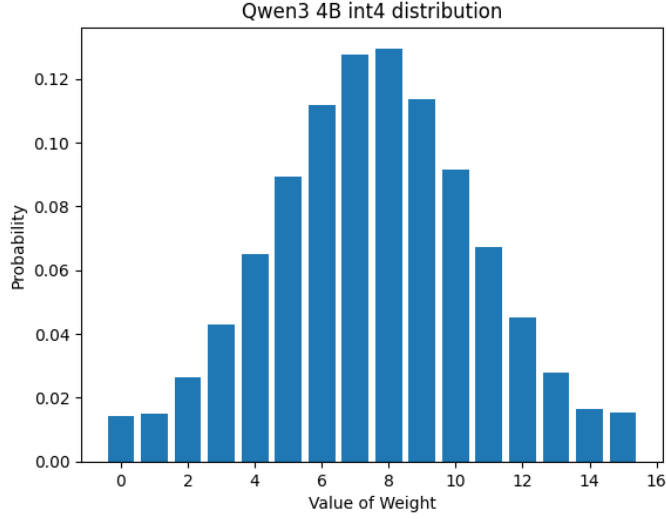


Fig. 2.3: Distribution of Qwen3-4B Weights (int4)

We notice that the full dynamic range of bfloat16 is actually not used. Especially in bfloat16, the exponents are clustered towards the center, with an average of 120 (this means most weights are $m \cdot 2^{-7}$). Several factors may contribute to this phenomenon: weight decay, regularization, and the optimizer (for example, stochastic gradient descent provides implicit optimization [28]).

We therefore conclude that large language models often have much room for compression, and is not using all of its potential capacity. This provides motivation for research in compression methods.

2.2 Small Language Models

In this section, we will examine TinyStories [29], a work that pushes the boundaries of small language models, and provide a few custom language models for further experimentation. By using a low-variance synthetic dataset (children’s stories) generated by GPT-3.5 and GPT-4 containing only words a typical 3-4 year old would understand, they were able to generate coherent output using models with as few as 1 million parameters.

One limitation we observed in this work is their usage of the default GPT-Neo [30] tokenizer (which is identical to the pretrained GPT-2 [31] tokenizer), restricted to the top 10,000 tokens. Using a generic tokenizer for a low-vocabulary dataset like TinyStories might result in wasted weights (tokens that never appear in TinyStories), and might not provide the optimal tokenization for our specialized dataset.

As a preliminary experiment, we train two small language models on TinyStories v1 and TinyStories v2, respectively. We use a SentencePiece Byte-Pair Encoding (BPE) tokenizer [32] with a vocabulary size of 1024, and we used a LLaMa-based [33] model with roughly half the parameters of the smallest reference model.

2.2.1 Evaluation: Quantitative Metric

We use perplexity [34] as a quantitative metric to evaluate a language model’s accuracy. For a model θ , a text x of length N , it is defined as the following.

$$\text{Perplexity}(\theta, x) = \left(\prod_{i=1}^N \frac{1}{p_{\theta}(x_i \mid x_1, x_2, \dots, x_{i-1})} \right)^{\frac{1}{N}} \quad (2.7)$$

In other words, it is the geometric mean of the inverse probabilities of a given text. To show this with an example, suppose we have a model θ and a ground truth text “Mary had a little lamb”. Suppose that the model’s predicted probabilities of this text is the following.

Table 2.3: Example Model Output Probabilities

Item	Value
$p_{\theta}(\text{“Mary”})$	0.5
$p_{\theta}(\text{“had”} \mid \text{“Mary”})$	0.7
$p_{\theta}(\text{“a”} \mid \text{“Mary”, “had”})$	0.9
$p_{\theta}(\text{“little”} \mid \text{“Mary”, “had”, “a”})$	0.6
$p_{\theta}(\text{“lamb”} \mid \text{“Mary”, “had”, “a”, “little”})$	0.9

Then the perplexity of the model on the text is the following.

$$\text{Perplexity} = \left(\frac{1}{0.5} \cdot \frac{1}{0.7} \cdot \frac{1}{0.9} \cdot \frac{1}{0.6} \cdot \frac{1}{0.9} \right)^{\frac{1}{5}} \quad (2.8)$$

$$\approx 1.49 \quad (2.9)$$

Intuitively, perplexity is a measure of how surprised or confused a model is when seeing a certain text. We compute the model’s perplexity over its validation data, then use the result as a metric to evaluate the model’s accuracy (lower is better).

2.2.2 Evaluation: Results and Discussion

For qualitative evaluation, we report sample outputs from our models on the prompt “Once upon a time, ”.

TinyStories V1 Model:

Once upon a time, there lived a little girl named Lily. Lily loved to play outside with her friends. One day, Lily saw a big pile of candy in the garden. She wanted to play too, but she wanted to play too. Lily went to her mom and said, “I want to play with my candy!” They both felt scared and didn’t want to play with her friends. So, they both went to get their candy. They had so much fun to play together and splashed until they got lost. They both had a great time playing together. They played together and had lots of fun.

TinyStories V2 Model:

Once upon a time, there was a little girl named Mia. Mia liked to play with her ball. She would throw it in the air and catch it. Mia was very happy. One day, Mia found a big box of balls on the ground. She picked it up and put it in the box. She picked it up and put it in the box. Mia put the ball back in the box and put it back on the ball. She was so happy! She played with the ball all day.

For quantitative evaluation, we report the validation perplexity of our two models. Note that the original paper only provided models trained on TinyStories v1.

Table 2.4: Evaluation of Our Language Models

	Reference	Ours (TinyStories v1)	Ours (TinyStories v2)
Architecture	GPT-Neo	LLaMa	LLaMa
Parameter Count	1.17M	0.57M	0.46M
Training Data	TinyStories v1		TinyStories v2
Tokenizer (Vocab Size)	GPT-Neo	SentencePiece	SentencePiece
Vocab Size	10000	1024	1024
Perplexity ↓ (TinyStories v1)	19.3919	11.9006	—
Perplexity ↓ (TinyStories v2)	—	—	8.0994

We observe that our model results in better perplexity than the reference model despite being much smaller. In `float32`, each weight is 4 bytes, so our TinyStories v1 model with 0.57M parameters is expected to take only ~ 2.17 MiB of disk space (the actual disk space, including the model’s configuration, is ~ 2.22 MiB). Models of this scale can be suitable for devices with extremely low compute and/or memory.

We will use our models as a base to conduct several more experiments.

2.3 Baseline Round-to-Nearest Quantization

The baseline for post-training quantization is symmetric uniform quantization using round-to-nearest (RTN) rounding [35]. A model can be quantized to b bits using Algorithm 1, then dequantized using Algorithm 2.

Algorithm 1 Baseline RTN Quantization**Require:** W (unquantized weights), b (target bit width)**Ensure:** Quantized model W_q

- 1: $q_{\min} \leftarrow -2^{b-1}$
- 2: $q_{\max} \leftarrow 2^{b-1} - 1$
- 3: $\text{scale} \leftarrow \frac{\max(W)}{q_{\max}}$
- 4: $W_q \leftarrow \text{clamp}(\lceil \frac{W}{\text{scale}} \rceil, q_{\min}, q_{\max})$
- 5: **return** W_q, scale

Algorithm 2 Baseline RTN Dequantization**Require:** W_q (quantized weights), scale **Ensure:** Dequantized weights $\hat{W}$

- 1: $\hat{W} \leftarrow W_q \cdot \text{scale}$
- 2: **return** $\hat{W}$

2.4 Quantization is Weight Perturbation

Let us establish the observation that quantization results in a perturbed model, using an example. Consider a model θ with parameters $W = \begin{bmatrix} 1.11 \\ 2.22 \\ 3.33 \end{bmatrix}$. By applying the baseline round-to-nearest quantization from Section 2.3, we obtain the following quantized model.

$$\text{scale} \approx 0.0262, W_q = \begin{bmatrix} 0b00101010 \\ 0b01010101 \\ 0b01111111 \end{bmatrix} = \begin{bmatrix} 42 \\ 85 \\ 127 \end{bmatrix} \quad (2.10)$$

After dequantization, we obtain the following model.

$$W_{\text{reconstructed}} = \text{scale} \cdot W_{\text{quantized}} \approx \begin{bmatrix} 1.1013 \\ 2.2287 \\ 3.33 \end{bmatrix} \quad (2.11)$$

We can see that the model is perturbed by the following.

$$\Delta W = \begin{bmatrix} 1.1013 \\ 2.2287 \\ 3.33 \end{bmatrix} - \begin{bmatrix} 1.11 \\ 2.22 \\ 3.33 \end{bmatrix} = \begin{bmatrix} -0.0087 \\ 0.0087 \\ 0 \end{bmatrix} \quad (2.12)$$

It is important to note the direction of perturbation. In this case, our model moved in the direction of $\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$.

2.5 Effects of Perturbation of Model Weights

As we established in Section 2.4, quantization perturbs weights in a certain direction. In this section, we examine the effects of weight perturbation using our TinyStories v2 model, by viewing its loss landscape.

2.5.1 Language Model Optimization Process

To understand our model’s loss landscape, we must first understand its loss function. The goal in optimizing language models is to minimize the difference between the ground truth and the model’s predictions. We measure this using the Kullback-Leibler (KL) divergence [36], defined as follows.

$$D_{KL}(P \parallel Q) = \sum_{x \in X} p_P(x) \ln \frac{p_P(x)}{p_Q(x)} \quad (2.13)$$

The KL-divergence measures how an approximating distribution Q is different from a distribution P . We use one-hot encoding to encode the ground truth into the distribution P , and use the model’s outputs as the approximating distribution Q .

Note that we once again need to employ the convention in Equation 2.5. Let us show why with an example: if we are predicting a word “Hello”, the ground truth P distribution and the predicted Q distribution might be:

Table 2.5: Example of KL-divergence

$\hat{y}$	$p_P(\hat{y})$	$p_Q(\hat{y})$
“Today”	0	0.07
“Hello”	1	0.81
“Cat”	0	0.10
“Rain”	0	0.02

In this case, the KL-divergence is:

$$D_{KL}(P \parallel Q) = \underbrace{0 \cdot \ln \frac{0}{0.07}}_{\text{“Today”}} + \underbrace{1 \cdot \ln \frac{1}{0.81}}_{\text{“Hello”}} + \underbrace{0 \cdot \ln \frac{0}{0.10}}_{\text{“Cat”}} + \underbrace{0 \cdot \ln \frac{0}{0.02}}_{\text{“Rain”}}, \quad (2.14)$$

which contains three undefined terms in the form of $0 \cdot \ln(\frac{0}{n})$. Using the convention from Equation 2.5, we can compute the undefined terms as follows.

$$0 \cdot \ln \frac{0}{n} = \underbrace{0 \ln 0}_{\text{0 by convention}} - 0 \ln n = 0 \quad (2.15)$$

We then obtain the following.

$$D_{KL}(P \parallel Q) = -\ln(0.81) \approx 0.21 \quad (2.16)$$

When minimizing the KL-divergence between the model predictions and the ground truth, we can use the following observation to ignore the constant $H(P)$ term, or the entropy of the ground truth distribution P .

$$D_{KL}(P \parallel Q) = \underbrace{\sum_{x \in X} p_P(x) \ln p_P(x)}_{-H(P)} - \sum_{x \in X} p_P(x) \ln p_Q(x) \quad (2.17)$$

$$\propto - \sum_{x \in X} p_P(x) \ln p_Q(x) \quad (2.18)$$

When optimizing, we only need to minimize $-\sum_{x \in X} p(x) \ln q(x)$. This is also known as the cross entropy $H(P, Q)$ between distributions P and Q .

The cross entropy for a language model θ on a single token, where P_i is the one-hot encoded ground truth and Q_i is the model's predicted distribution, is the following.

$$H(P_i, Q_i) = - \sum_{x \in X} p_{P_i}(x) \ln p_{Q_i}(x) = -\ln p_\theta(x_i \mid x_{<i}) \quad (2.19)$$

The loss function, the average cross-entropy per token over a sequence of length N , is the following.

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^N H(P_i, Q_i) = -\frac{1}{N} \sum_{i=1}^N \ln p_\theta(x_i \mid x_{<i}) \quad (2.20)$$

Note that this is exactly the natural logarithm of the perplexity of the model on the given text. Given this loss function, we will proceed to investigate the loss landscape of our model.

2.5.2 Loss Landscape

Since the loss landscape of a language model is in n -dimension, where n is its number of parameters, it is difficult to be directly visualized. We will investigate the loss landscape of our TinyStories v2 model from Section 2.2 by perturbing its weights and computing the perplexity of the perturbed models.

Straight Line

First, we will sample models along a simple line in the following way.

$$\theta' = \theta + \epsilon, \epsilon \in \{-0.0020 \dots 0.0015\} \quad (2.21)$$

Note that the blue line in the figure represents $\epsilon \in \{-0.0015 \dots 0.0015\}$. From the plot in Figure 2.4, we notice that there is a “basin” of low loss, and that moving towards the negative direction caused less degradation than moving towards the positive direction.

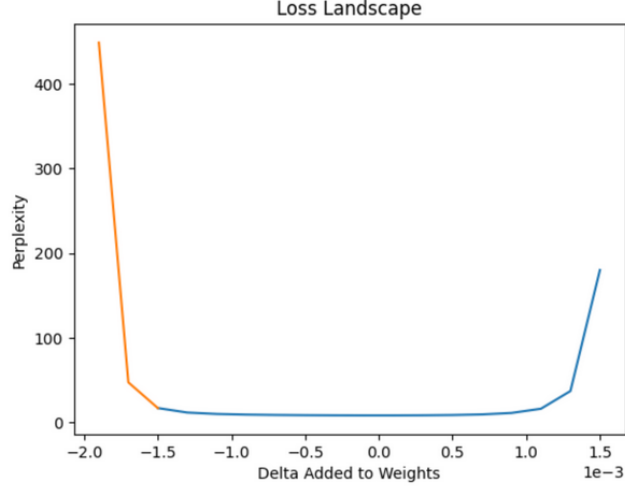


Fig. 2.4: Models Along a Line

Moving Randomly

Next, we sample models stochastically from the following distribution. We sample 5 models for each σ .

$$\theta' \sim \mathcal{N}(\theta, \sigma^2 \mathbf{I}), \sigma^2 \in \{0.0005 \dots 0.0040\} \quad (2.22)$$

Note that we cap the perplexity at 100 in Figure 2.5 for better visualization, because some perplexities are extremely large.

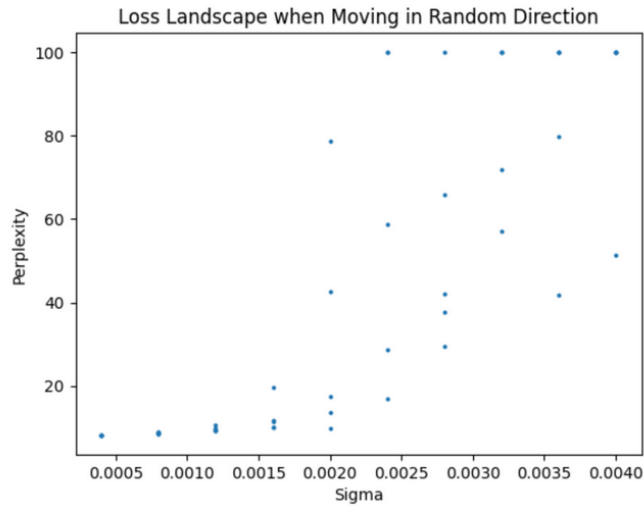


Fig. 2.5: Models from Gaussian Distribution

In Figure 2.6a, we plot the average perplexities of each σ . We notice an outlier at $\sigma = 0.0024$, and removing it yields Figure 2.6b.

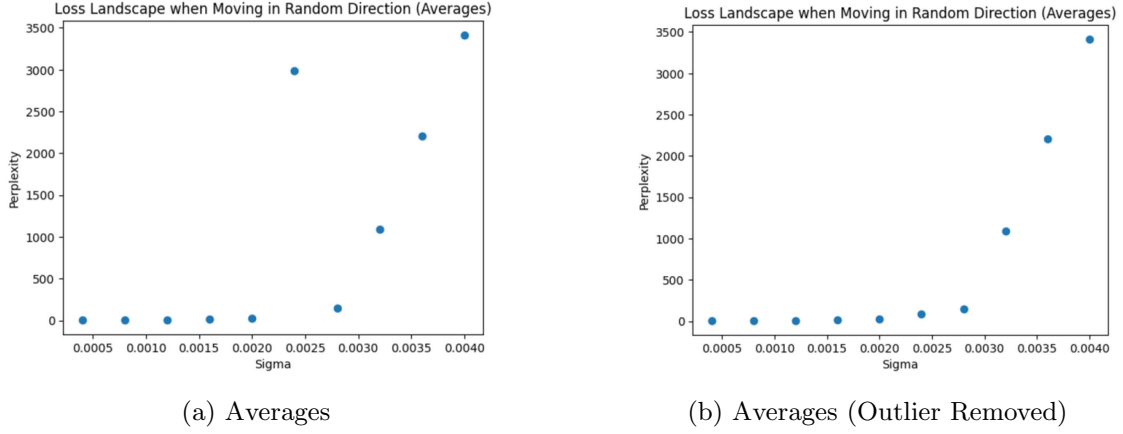


Fig. 2.6: Loss Landscape of Models from Gaussian Distribution

This further confirms the observation that when perturbing a model, the direction of perturbation is important, and some directions yield terrible results (like the outlier) while some directions preserve the model's accuracy more.

Towards Another Model

We pretrain another model with the same configuration as the base model, and move from the base model to the other model. The distance between these models, $\|\theta - \theta'\|_2$, is 75.6, which means they're quite far apart, and the magnitude of perturbation is much greater than the previous experiments.

Note that when the x-axis is 0, the model is exactly the base model, and when the x-axis is 1, the model is exactly the other pretrained model.

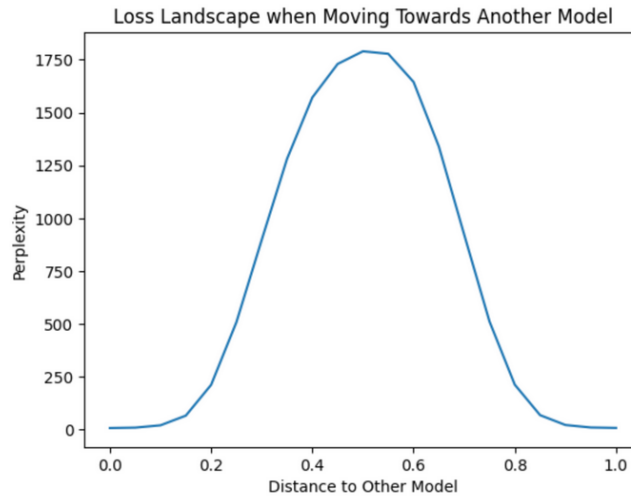


Fig. 2.7: Moving Towards Another Model

We observe from Figure 2.7 that even though the magnitude of perturbation is much greater than the previous experiments, the increase in loss is not as significant. This shows that the model is less sensitive to perturbations in this direction.

2.5.3 Discussion

The key insight from this experiment is the fact that when weights of a model are perturbed, the loss does not degrade uniformly. And there exist directions of perturbation that result in massive loss increases, which outline the potential need for optimizing the direction of rounding in post-training quantization.

Chapter 3

Literature Review

3.1 AdaRound

3.1.1 Notation

Let us establish the notation for variables in Table 3.1.

Table 3.1: Notation for this Literature Review

Symbol	Description
w_j	Weights of a layer j
x_j	Input to a layer j
b	Target bit-width quantization
scale	Continuous scaling factor
$\lfloor \cdot \rfloor$	Round-to-nearest-integer operator
$\lfloor \cdot \rfloor$	Floor operator
$\mathbf{W}, \hat{\mathbf{W}}$	Matrices of unquantized and quantized weights
$\mathcal{L}$	A loss function

3.1.2 Introduction to AdaRound

AdaRound [20] is a novel rounding method, based on the key insight that round-to-nearest is suboptimal for post-training quantization (PTQ).

The authors' proposed method is to instead minimize the difference in task loss of the quantized model and the unquantized model. The task loss of layer j of the unquantized model is $\mathcal{L}(x, y, w_j)$, and the task loss of the same layer of the quantized model $\mathcal{L}(x, y, w_j + \Delta w_j)$. Therefore, the authors' objective is the following.

$$\arg \min_{\Delta w_j} (\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j)) \quad (3.1)$$

The authors use a second order Taylor expansion to approximate the above objective.

$$\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j) \approx \nabla \mathcal{L}(x, y, w_j)^T \Delta w_j + \frac{1}{2} \Delta w_j^T \mathbf{H} \Delta w_j \quad (3.2)$$

Where $\mathbf{H}$ is the Hessian $\nabla^2 \mathcal{L}(x, y, w_j)$. The authors then ignore the first term by asserting that the gradient $\nabla \mathcal{L}(x, y, w_j)$ is 0 for converged pretrained models, which simplifies into the following.

$$\mathcal{L}(x, y, w_j + \Delta w_j) - \mathcal{L}(x, y, w_j) \approx \frac{1}{2} \Delta w_j^T \mathbf{H} \Delta w_j \quad (3.3)$$

Finally, the authors formulate the optimization of weigh rounding for layer j as the following Quadratic Unconstrained Binary Optimization (QUBO) problem.

$$\arg \min_{\Delta w_j} \Delta w_j^T \mathbf{H} \Delta w_j \quad (3.4)$$

The above problem is NP-hard. To solve this efficiently, AdaRound uses the local mean squared error alongside a regularizer as a proxy. The regularizer encourages the optimization variables to converge towards either 0 or 1.

$$\arg \min_{\mathbf{V}} \left(\|\mathbf{W}x - \hat{\mathbf{W}}x\|_F^2 + f_{\text{reg}}(\mathbf{V}) \right) \quad (3.5)$$

The regularizer uses an annealed parameter β and is defined as follows.

$$f_{\text{reg}}(\mathbf{V}) = \sum_{i,j} 1 - |2h(\mathbf{V}_{i,j}) - 1|^\beta \quad (3.6)$$

The above objective is then optimized over

$$\hat{\mathbf{W}} = \text{scale} \cdot \text{clamp} \left(\left\lfloor \frac{\mathbf{W}}{\text{scale}} \right\rfloor + h(\mathbf{V}), q_{\min}, q_{\max} \right), \quad (3.7)$$

$$q_{\min} = -2^{b-1}, \quad (3.8)$$

$$q_{\max} = 2^{b-1} - 1. \quad (3.9)$$

The authors define h as a rectified sigmoid as follows, although it can be any differentiable function with a range of $h(\mathbf{V}_{i,j}) \in [0, 1]$.

$$h(\mathbf{V}_{i,j}) = \text{clamp}(\sigma(\mathbf{V}_{i,j})(\zeta - \gamma) + \gamma, 0, 1), \quad (3.10)$$

$$\zeta = 1.1, \quad (3.11)$$

$$\gamma = -0.1. \quad (3.12)$$

The use of a rectified sigmoid is justified as it has non-vanishing gradients near 0 and 1, and helps the learning process as they previously used a regularizer to encourage $h(\mathbf{V}_{i,j})$ to move to either 0 or 1.

3.1.3 AdaRound Algorithm

We implement AdaRound in Algorithm 3.

Algorithm 3 AdaRound

Require: $\mathbf{W}$, X (calibration data), t (number of iterations), b (target bit width), β (regularization parameter)

Ensure: Quantized model $\mathbf{W}_q$, scales

- 1: $x_{in} \leftarrow \text{CACHEACTIVATIONS}(\mathbf{W}, X)$ ▷ Forward pass on calibration data to cache input activations
 - 2: **for** w_j in $\mathbf{W}$ **do**
 - 3: $q_{\min} \leftarrow -2^{b-1}$
 - 4: $q_{\max} \leftarrow 2^{b-1} - 1$
 - 5: $\text{scales}_j \leftarrow \frac{\max(w_j)}{q_{\max}}$ ▷ Store scale for output
 - 6: $\mathbf{V}_j \leftarrow h^{-1}(w_j - \lfloor w_j \rfloor)$ ▷ Initialize $\mathbf{V}_j$ so that it corresponds to the fraction of each weight
 - 7: **for** $i \in \{0 \dots t\}$ **do**
 - 8: $\beta \leftarrow \text{ANNEAL}(\beta)$
 - 9: $\hat{w}_j \leftarrow \text{clamp}\left(\lfloor \frac{w_j}{\text{scales}_j} \rfloor + h(\mathbf{V}_j), q_{\min}, q_{\max}\right)$
 - 10: $\mathcal{L} \leftarrow \|w_j x_{in} - \hat{w}_j x_{in}\|_F^2 + \sum_{i,j} (1 - |2h(\mathbf{V}_{i,j}) - 1|^\beta)$
 - 11: $\mathbf{V} \leftarrow \mathbf{V} - \nabla \mathcal{L}$ ▷ via autograd
 - 12: **end for**
 - 13: $\mathbf{W}_{Q_{i,j}} \leftarrow \begin{cases} \lceil w_{i,j} \rceil, & \text{where } h(\mathbf{V}_{i,j}) \geq 0.5 \\ \lfloor w_{i,j} \rfloor, & \text{otherwise} \end{cases}$ ▷ Finalize quantization
 - 14: **end for**
 - 15: **return** $\mathbf{W}_Q$, scales
-

3.1.4 Limitations

We identify two important limitations of AdaRound. First, using calibration data can cause some information to be encoded into the weight perturbation matrix. We empirically verify this by demonstrating a novel inversion method to reconstruct representations of training data in Chapter 4. Second, the assumption in Equation 3.2 that the gradient of the loss function for pretrained converged models is negligible (for which empirical evidence was not provided) is challenged by recent literature, which we review below.

3.2 Edge of Stability and Central Flows

In this section, we review a rebuttal to the zero-gradient assumption in Equation 3.2.

Consider gradient descent [37], where we compute the gradient of the loss function then try to move the model weights in the opposite direction to minimize the loss.

$$\theta_{t+1} = \theta_t - \eta \nabla \mathcal{L}(\theta) \quad (3.13)$$

While one might expect weights to follow a smooth, stable path to an optimum, gradient descent with deep neural networks almost always operate in a highly dynamic, oscillatory regime described as the **Edge of Stability** [38], leading to chaotic loss curves such as the follows.

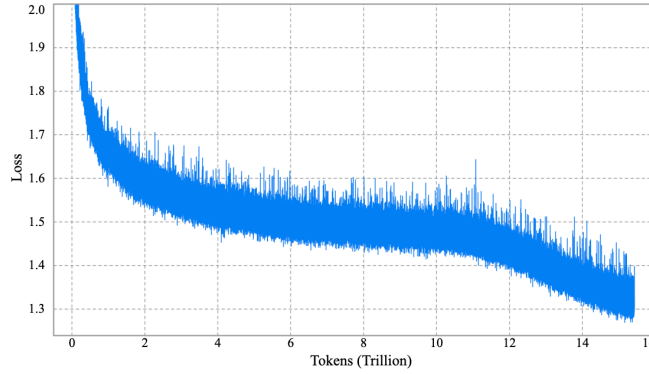


Fig. 3.1: Training Loss Curve of Kimi K2 [39]

It can be observed that the oscillations in gradient descent can be viewed as bouncing between walls of a valley of low gradient [40]. **Central Flows** [21] are differential equations that attempts to model the time-averaged trajectory of an oscillatory optimizer, illustrated as follows.

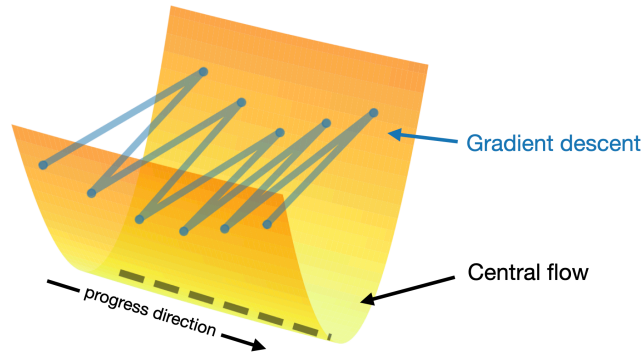


Fig. 3.2: Oscillatory Optimizer Bouncing between Walls in a Valley [21]

The oscillatory optimization also implies that the gradient for pretrained models are rarely zero. The authors empirically show that even as the training loss decreases, the gradient of a model almost stays constant, as seen in Figure 3.3.

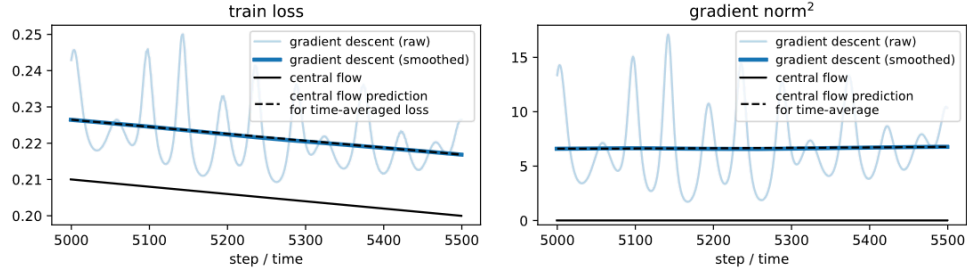


Fig. 3.3: Gradient Norms in Vision Transformer Trained on CIFAR10 [21, pg. 15]

This observation creates a conflict with AdaRound’s assumption in Equation 3.2 that the gradient for converged pretrained models is zero, which we attempt to reconcile in Chapter 5.

Chapter 4

AdaRound Inversion Algorithm

In this chapter, we propose and evaluate a novel algorithm to reconstruct representations of training data in AdaRound.

4.1 Formulation of Leakage Channel

AdaRound’s objective function is expected to approach zero after training. Let us define the weight perturbation matrix as $\Delta W = W - \hat{W}$. The objective function can be rewritten as the following.

$$\|Wx - \hat{W}x\|_F^2 = \|\Delta Wx\|_F^2 \quad (4.1)$$

After optimization, we expect the following.

$$\|\Delta Wx\|_F^2 \approx 0 \iff \Delta Wx \approx 0 \quad (4.2)$$

We notice that $\Delta Wx = 0 \implies x \in \mathcal{N}(\Delta W)$. Therefore, the training data x used to calibrate the AdaRound process approximately lies within the null space of the weight perturbation matrix ΔW .

4.2 Method

An attacker with access to the original weights W and the quantized weights $\hat{W}$ can generate a sample $\hat{x}$ that lies within this null space. We can initialize a $\hat{x}$ from random noise and apply Stochastic Gradient Descent with the objective $\arg \min_{\hat{x}} \|\Delta W\hat{x}\|_2^2$, as described in Algorithm 4.

Algorithm 4 Proposed Inversion Method**Require:** $W, \hat{W}, \hat{x}_{\text{init}}$, Number of Iterations K , Learning Rate η

-
- 1: $\Delta W \leftarrow W - \hat{W}$
 - 2: $\hat{x} \leftarrow \hat{x}_{\text{init}}$
 - 3: **for** $k = 1$ **to** K **do**
 - 4: $\mathcal{L} \leftarrow \sum (\Delta W \tanh(\hat{x}))^2$ (Forward pass)
 - 5: $\frac{\partial \mathcal{L}}{\partial \hat{x}} \leftarrow \text{Autograd}(\mathcal{L}, \hat{x})$ (Backward pass via autograd [41])
 - 6: $\hat{x} \leftarrow \hat{x} - \eta \frac{\partial \mathcal{L}}{\partial \hat{x}}$ (Update $\hat{x}$)
 - 7: **end for**
 - 8: **return** $\tanh(\hat{x})$
-

Note that the objective can be rewritten as the following.

$$\|\Delta W \hat{x}\|_2^2 = (\Delta W \hat{x})^T (\Delta W \hat{x}) = \hat{x}^T (\Delta W^T \Delta W) \hat{x} \quad (4.3)$$

A simple optimization can therefore be made by computing the Gram matrix $G = \Delta W^T \Delta W$ once, then use it to calculate the objective as $\hat{x}^T G \hat{x}$ in each iteration. (See Algorithm 5)

Algorithm 5 Proposed Inversion with Optimization**Require:** $W, \hat{W}, \hat{x}_{\text{init}}$, Number of Iterations K , Learning Rate η

-
- 1: $\Delta W \leftarrow W - \hat{W}$
 - 2: $\hat{x} \leftarrow \hat{x}_{\text{init}}$
 - 3: $G \leftarrow \Delta W^T \Delta W$
 - 4: **for** $k = 1$ **to** K **do**
 - 5: $\mathcal{L} \leftarrow \sum (\tanh(\hat{x})^T G \tanh(\hat{x}))$ (Forward pass)
 - 6: $\frac{\partial \mathcal{L}}{\partial \hat{x}} \leftarrow \text{Autograd}(\mathcal{L}, \hat{x})$ (Backward pass via autograd [41])
 - 7: $\hat{x} \leftarrow \hat{x} - \eta \frac{\partial \mathcal{L}}{\partial \hat{x}}$ (Update $\hat{x}$)
 - 8: **end for**
 - 9: **return** $\tanh(\hat{x})$
-

4.3 Evaluation

4.3.1 Experiment Setup

We trained MLPs on these datasets: Olivetti Faces [42], Chest X-Ray Images (Pneumonia) [43], and the Weizmann Horse Database [44], on a single computer (CPU: Intel Xeon w9-3495X, GPU: 1x NVIDIA GeForce RTX 4090). We perform 500 iterations of AdaRound quantization using the Adam optimizer, then obtain ΔW , the difference between the quantized and the original model. For each model, we generate 10 random samples from a Gaussian distribution, then train for 3000 iterations using the Adam optimizer. We report the SSIM (Structural Similarity Index Measure), PSNR (Peak Signal-to-Noise ratio), and Normalized RMSE (Root Mean Squared Error) between the mean of the calibration data and the mean of 10 generated samples before and after our inversion

method in Table 4.2. We also report a visualization of the mean calibration sample vs. the mean generated sample in Figures 4.1, 4.2, and 4.3.

Our implementation details are described in Table 4.1.

Table 4.1: Implementation Details

Hyperparameter/Component	Value
Number of Iterations	3000
Learning Rate	0.05
Optimizer	Adam [45]
Learning Rate Scheduler	CosineAnnealingLR [46]

4.3.2 Results

Table 4.2: Metrics from Random Initialization vs. Our Proposed Method

	SSIM $\uparrow$	PSNR $\uparrow$	NRMSE $\downarrow$
Olivetti Faces (Before)	0.00	3.50	1.16
Olivetti Faces (After)	0.14 (+0.14)	12.83 (+9.33)	0.40 (-0.76)
Chest X-Ray (Before)	0.00	2.61	1.10
Chest X-Ray (After)	0.07 (+0.07)	10.13 (+7.52)	0.46 (-0.64)
Horse DB (Before)	0.00	2.60	1.11
Horse DB (After)	0.07 (+0.07)	10.15 (+7.55)	0.47 (-0.64)

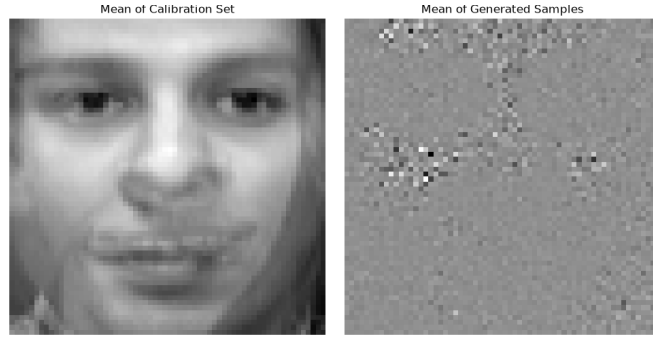


Fig. 4.1: Results on Olivetti Faces

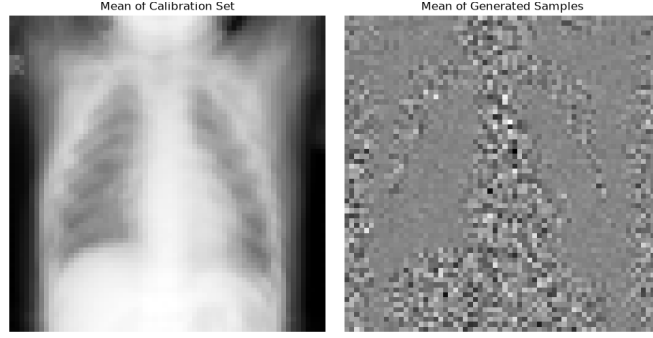


Fig. 4.2: Results on Chest X-Ray Images (Pneumonia)

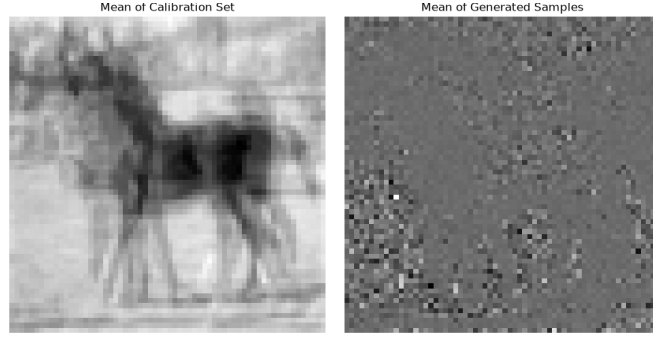


Fig. 4.3: Results on Weizmann Horse Database

We notice an increase in SSIM and PSNR, and a decrease in Normalized RMSE, implying that the mean of the generated samples resembles the mean of the calibration samples to some extent.

4.4 Ablation Study

We perform an ablation study by applying our inversion method to round-to-nearest (RTN) quantization. Since RTN does not use calibration data, we compare our method’s samples to the mean of the training data instead. We report the same metrics as the main experiment (SSIM, PSNR, and Normalized RMSE) in Table 4.3 and a visualization in Figures 4.4, 4.5, and 4.6.

Table 4.3: Results from Ablation Study

	SSIM $\uparrow$	PSNR $\uparrow$	NRMSE $\downarrow$
Olivetti Faces (Before)	0.00	2.89	1.12
Olivetti Faces (After)	0.00	3.90 (+1.01)	1.00 (-0.12)
Chest X-Ray (Before)	0.00	2.77	1.11
Chest X-Ray (After)	0.00	3.64 (+0.87)	1.00 (-0.11)
Horse DB (Before)	0.00	2.57	1.10
Horse DB (After)	0.00	3.38 (+0.81)	1.00 (-0.10)

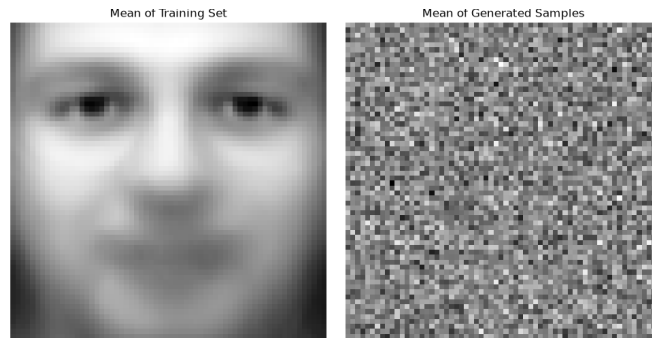


Fig. 4.4: Ablation Study Results on Olivetti Faces

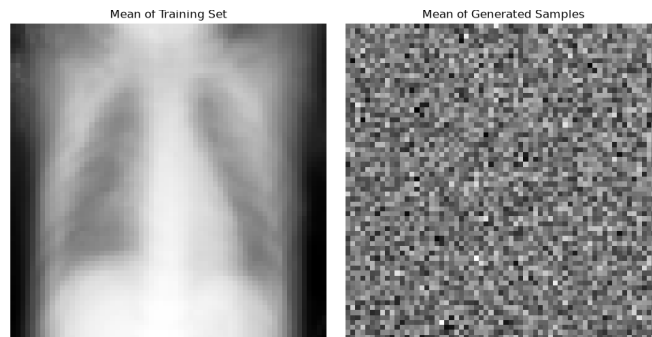


Fig. 4.5: Ablation Study Results on Chest X-Ray Images (Pneumonia)

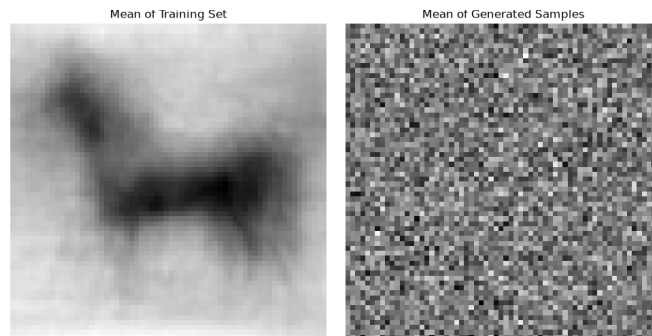


Fig. 4.6: Ablation Study Results on Weizmann Horse Database

We notice 0.00 SSIM on all datasets, implying that the generated samples do not look like the calibration data at all.

4.5 Discussion

4.5.1 Main Experiment

From the metrics, we observe an increase in SSIM and PSNR, and a decrease in Normalized RMSE after training. In the images, we can see the silhouettes and outlines of

the calibration set, although it is quite difficult to make out. From the results, we can conclude that the ΔW of a model quantized using AdaRound does contain information about the calibration data.

4.5.2 Ablation Study

Interestingly, we see a slight improvement in PSNR and NRMSE in the ablation study. However, crucially the SSIM shows that round-to-nearest quantization is not vulnerable to our method. In other words, our method indeed extracts information about the calibration data in data-dependent quantization methods.

4.5.3 Overall

We demonstrate that AdaRound encodes information from the calibration data into the weight perturbation matrix. By targeting the null space of this matrix, our proposed method generates samples that resembles outlines of the calibration data. While the reconstructed data only represents the mean of the calibration data rather than exact instances, the leakage is significant enough to warrant reconsideration of the security of data-dependent PTQ methods.

4.5.4 Limitations

The most important limitation is the fact that this method only reconstructs representations of the mean of the calibration data, but not individual samples. The low SSIM (~ 0.14) is another limitation: this method only shows evidence for the leakage channel but the quality may be improved. For future work, it is possible to consider adding Total Variation (TV) loss [47], or using a prior such as a deep image prior [48], to improve the quality of the reconstructed samples.

Chapter 5

Gradient-Aware Adaptive Rounding

In this chapter, we propose and evaluate a gradient-aware adaptive rounding method.

5.1 Method

To reconcile AdaRound’s assumption that the gradient term in Equation 3.2 can be ignored, and Central Flows’ evidence that the gradient is nonzero even for pretrained converged models, we propose a modified version of AdaRound to incorporate awareness of the local gradient.

In order to minimize

$$\mathbb{E} \left[g^{(w)^T} \Delta W \right], \quad (5.1)$$

we can attempt to force ΔW to have the same sign as $-g^{(w)^T}$. We approximate this by finetuning a model on a small amount of training data, and compute the Exponential Moving Average (EMA) of the model during each iteration. We then use $\text{sign}(W'_{\text{avg}} - W)$, the sign of the difference between the weights before and after finetuning, as a proxy to $-\text{sign}(g^{(w)^T})$. Finally, we add the Binary Cross Entropy (BCE) between the current rounding directions and $\text{sign}(W'_{\text{avg}} - W)$ as a penalty term to AdaRound’s optimization objective.

In other words, this is essentially a modification to AdaRound’s objective in Equation 3.5. Our version is the following.

$$\arg \min_{\mathbf{V}} \left(\|\mathbf{W}x - \hat{\mathbf{W}}x\|_F^2 + f_{\text{reg}}(\mathbf{V}) + \underbrace{\psi \cdot \text{BCE}(h(\mathbf{V}_j), \text{sign}(\mathbf{W}' - \mathbf{W}))}_{\text{Proposed addition}} \right) \quad (5.2)$$

Algorithm 6 Gradient-Aware Adaptive Rounding (Finetuning)

Require: $\mathbf{W}$, X (calibration data), η (learning rate), B (batch size)
Ensure: EMA of models during finetuning $\mathbf{W}'_{\text{avg}}$

```

1:  $\mathbf{W}' \leftarrow \mathbf{W}$ 
2:  $\mathbf{W}'_{\text{avg}} \leftarrow \mathbf{W}$ 
3:  $N \leftarrow \frac{X}{B}$  ▷ Number of Iterations
4:  $\alpha \leftarrow \frac{2}{N+1}$  ▷ EMA Smoothing Factor
5: for each batch  $x_b$  in  $X$  do
6:    $\hat{x}_b \leftarrow \mathbf{W}'(x_b)$  ▷ Forward pass
7:    $\mathcal{L} \leftarrow \text{CrossEntropyLoss}(\hat{x}_b, x_b)$ 
8:    $\mathbf{W}' \leftarrow \mathbf{W}' - \eta \cdot \nabla \mathcal{L}$  ▷ Backward pass
9:    $\mathbf{W}'_{\text{avg}} \leftarrow \alpha \cdot \mathbf{W}'_{\text{avg}} + (1 - \alpha) \cdot \mathbf{W}'$  ▷ Update EMA
10: end for
11: return  $\mathbf{W}'_{\text{avg}}$ 

```

Algorithm 7 Gradient-Aware Adaptive Rounding (Quantization)

Require: $\mathbf{W}$, $\mathbf{W}'_{\text{avg}}$, X (calibration data), t (number of iterations), b (target bit width), β (regularization parameter), ψ (BCE loss multiplier)
Ensure: Quantized model $\mathbf{W}_Q$, scales

```

1:  $x_{in} \leftarrow \text{CACHEACTIVATIONS}(\mathbf{W}, X)$  ▷ Forward pass on calibration data to cache input activations
2: for  $w_j$  in  $\mathbf{W}$  do
3:    $\mathbf{D} \leftarrow \text{SIGN}(\mathbf{W}'_{\text{avg}} - \mathbf{W})$ 
4:    $q_{\min} \leftarrow -2^{b-1}$ 
5:    $q_{\max} \leftarrow 2^{b-1} - 1$ 
6:    $\text{scales}_j \leftarrow \frac{\max(w_j)}{q_{\max}}$  ▷ Store scale for output
7:    $\mathbf{V}_j \leftarrow h^{-1}(w_j - \lfloor w_j \rfloor)$  ▷ Initialize  $\mathbf{V}_j$  so that it corresponds to the fraction of each weight
8:   for  $i \in \{0 \dots t\}$  do
9:      $\beta \leftarrow \text{ANNEAL}(\beta)$ 
10:     $\hat{w}_j \leftarrow \text{clamp}\left(\lfloor \frac{w_j}{\text{scales}_j} \rfloor + h(\mathbf{V}_j), q_{\min}, q_{\max}\right)$ 
11:     $\mathcal{L} \leftarrow \underbrace{\psi \cdot \text{BCE}(h(\mathbf{V}_j), \mathbf{D})}_{\text{Proposed addition}} + \|w_j x_{in} - \hat{w}_j x_{in}\|_F^2 + \sum_{i,j} (1 - |2h(\mathbf{V}_{i,j}) - 1|^\beta)$ 
12:     $\mathbf{V} \leftarrow \mathbf{V} - \nabla \mathcal{L}$  ▷ via autograd
13:   end for
14:    $\mathbf{W}_{Q_{i,j}} \leftarrow \begin{cases} \lceil w_{i,j} \rceil, & \text{where } h(\mathbf{V}_{i,j}) \geq 0.5 \\ \lfloor w_{i,j} \rfloor, & \text{otherwise} \end{cases}$  ▷ Finalize quantization
15: end for
16: return  $\mathbf{W}_Q$ , scales

```

5.2 Evaluation

In order to evaluate whether incorporating local gradient information can improve AdaRound quantization, we evaluate our proposed method on our two language models

from Section 2.2, and used perplexity as a metric to the models’ accuracy.

5.2.1 Experiment Setup

We evaluate the proposed method on the same computing environment as described in Section 4.3. We report implementation details for such as our hyperparameters in Table 5.1 (note that they are the same for TinyStories v1 and TinyStories v2). We report the perplexity of our two language models (TinyStories v1 and TinyStories v2) in Table 5.2.

Table 5.1: Implementation Details

Hyperparameter/Component	Value
Finetuning Data	2% of Training Data
Learning Rate	3e-4
Optimizer	AdamW [49]
Learning Rate Scheduler	CosineAnnealingLR [46]
ψ	100.0

5.2.2 Results

We observe neutral to negative results using the proposed method. It is important to note that the two models had different testing data (TinyStories v1 and TinyStories v2, respectively).

Table 5.2: Perplexity Evaluation of Proposed Method

Type of Quantization	TinyStories V1	TinyStories V2
None (Unquantized Model)	11.9006	8.0994
Round-to-Nearest	11.9219	8.1224
AdaRound	11.9216	8.1177
Proposed Method	11.9216	8.1179

5.3 Ablation Study

We report an ablation study by ablating two possible elements of the proposed method: AdaRound and EMA. Ablating EMA means we use final finetuned model weights instead of the EMA of model weights during all iterations. Ablating AdaRound means we steer the rounding using only the sign of the weight perturbation matrix, as described in Algorithm 8.

Table 5.3: Perplexity Evaluation of Ablation Studies

	AdaRound + BCE	EMA	TinyStories V1	TinyStories V2
Main Experiment	✓	✓	11.9216	8.1179
Ablation 1	✓	×	11.9216	8.1197
Ablation 2	×	✓	11.9166	8.1546
Ablation 3	×	×	11.9050	8.1084

Algorithm 8 Gradient-based Steering without AdaRound

Require: $\mathbf{W}$ (unquantized weights), $\mathbf{W}'$ (calibration model), b (target bit width)

Ensure: Quantized model $\mathbf{W}_q$

```

1: function CUSTOMROUND( $\mathbf{W}, \mathbf{W}'$ )
2:    $\mathbf{W}_{i,j} \leftarrow \begin{cases} \lceil \mathbf{W}_{i,j} \rceil, & \text{where } \mathbf{W}_{i,j} < \mathbf{W}'_{i,j} \\ \lfloor \mathbf{W}_{i,j} \rfloor, & \text{otherwise} \end{cases}$ 
3:   return  $\mathbf{W}$ 
4: end function
5:  $q_{\min} \leftarrow -2^{b-1}$ 
6:  $q_{\max} \leftarrow 2^{b-1} - 1$ 
7:  $\text{scale} \leftarrow \frac{\max(\mathbf{W})}{q_{\max}}$ 
8:  $\mathbf{W}_q \leftarrow \text{clamp}\left(\text{CUSTOMROUND}\left(\frac{\mathbf{W}}{\text{scale}}\right), q_{\min}, q_{\max}\right)$ 
9: return  $\mathbf{W}_q, \text{scale}$ 

```

5.4 Discussion

First, we notice neutral to negative results in the main experiment. While the central flows paper does suggest non-zero gradients even for pretrained converged models, adding this signal to the rounding did not result in a meaningful improvement.

We also notice that in our ablation study, we achieved the best performance by ablating both EMA and AdaRound (outperforming original AdaRound). This method steers the model during rounding in the direction to a finetuned model directly, and this ablation study shows that this direction can still potentially be useful despite the neutral to negative results in the main experiment. For example, it is possible that the EMA averaging smoothed out important signals, or BCE loss was not optimal for steering the model during rounding.

However, it is also possible that while the gradient is non-zero, the assertion that it is zero is simply “good enough” for the search of the optimal rounding direction.

Finally, it might be possible to consider a better proxy than the direction towards a finetuned model, as this is dependent on the finetuning hyperparameters (learning rate, step count, etc.).

Chapter 6

Conclusion

Given the background of the exponential scaling of Large Language Models (LLMs), this thesis first quantified the compressibility of modern LLMs and confirmed that their capacity is underutilized. We analysed the loss landscape of a language model to show the effects of weight perturbation direction on a model’s accuracy.

We review AdaRound, an existing method for superior weight rounding in post-training quantization. We find that it inadvertently encodes information about its calibration data into its weight perturbation matrix and empirically confirmed it with an inversion attack, albeit only reconstructing coarse structures with low SSIM.

Furthermore, we challenged an assertion in AdaRound that the gradient of the loss function for pretrained converged models is negligible. We used the direction toward a finetuned model as a proxy for the gradient direction, and incorporate that signal into AdaRound rounding. However, we obtained neutral to negative results using our proposed method.

In the future, we would like to improve the quality of the inversion attack using regularizers and priors. We would like to furthermore explore the incorporation of the gradient of the loss function in AdaRound, possibly using other proxies to estimate the direction of the gradient. Finally, we would like to replicate our experiments on models of larger scale, so we can determine whether its behavior persists beyond smaller on-device models.

Acknowledgements

First, I am deeply honored to be a part of the Nakamura-Takase laboratory. I would like to express my sincere gratitude to Professor Hiroshi Nakamura and Associate Professor Hideki Takase for their extensive and dedicated guidance throughout my studies. Without their support, this work would not have been possible, and I truly appreciate the time and energy they invested in me throughout my Master's program.

I would like to extend my thanks to all members of the Nakamura-Takase laboratory for their support and collaborative spirit. In particular, I am grateful to Dr. Takuya Kojima for managing the computing infrastructure used in this research, and to my lab tutor, Li Huilin, for his kind assistance during my transition into the laboratory. My thanks also go out to my friends and family — throughout my Master's program, they have also been supporting me kindly.

Finally, I gratefully acknowledge the financial support provided by the MEXT Scholarship throughout my Master's program.

Bibliography

- [1] Johannes Gerstmayr, Peter Manzl, and Michael Pieber. Multibody models generated from natural language. *Multibody System Dynamics*, Vol. 62, pp. 249–271, January 2024.
- [2] David Patterson, Joseph Gonzalez, Quoc Le, Chen Liang, Lluís-Miquel Munguia, Daniel Rothchild, David So, Maud Texier, and Jeff Dean. Carbon Emissions and Large Neural Network Training, April 2021.
- [3] Zahida Kausar, Seemab Latif, Raja Khurram Shahzad, and Mehwish Fatima. Quantifying the Climate Risk of Generative AI: Region-Aware Carbon Accounting with G-TRACE and the AI Sustainability Pyramid, May 2026.
- [4] Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten, Alex Vaughan, Amy Yang, Angela Fan, Anirudh Goyal, Anthony Hartshorn, Aobo Yang, Archi Mitra, Archie Sravankumar, Artem Korenev, Arthur Hinsvark, Arun Rao, Aston Zhang, Aurelien Rodriguez, Austen Gregerson, Ava Spataru, Baptiste Roziere, Bethany Biron, Binh Tang, Bobbie Chern, Charlotte Caucheteux, Chaya Nayak, Chloe Bi, Chris Marra, Chris McConnell, Christian Keller, Christophe Touret, Chunyang Wu, Corinne Wong, Cristian Canton Ferrer, Cyrus Nikolaidis, Damien Allonsius, Daniel Song, Danielle Pintz, Danny Livshits, Danny Wyatt, David Esiobu, Dhruv Choudhary, Dhruv Mahajan, Diego Garcia-Olano, Diego Perino, Dieuwke Hupkes, Egor Lakomkin, Ehab AlBadawy, Elina Lobanova, Emily Dinan, Eric Michael Smith, Filip Radenovic, Francisco Guzmán, Frank Zhang, Gabriel Synnaeve, Gabrielle Lee, Georgia Lewis Anderson, Govind Thattai, Graeme Nail, Gregoire Mialon, Guan Pang, Guillem Cucurell, Hailey Nguyen, Hannah Korevaar, Hu Xu, Hugo Touvron, Iliyan Zarov, Imanol Arrieta Ibarra, Isabel Kloumann, Ishan Misra, Ivan Evtimov, Jack Zhang, Jade Copet, Jaewon Lee, Jan Geffert, Jana Vranes, Jason Park, Jay Mahadeokar, Jeet Shah, Jelier van der Linde, Jennifer Billock, Jenny Hong, Jenya Lee, Jeremy Fu, Jianfeng Chi, Jianyu Huang, Jiawen Liu, Jie Wang, Jiecao Yu, Joanna Bitton, Joe Spisak, Jongsoo Park, Joseph Rocca, Joshua Johnstun, Joshua Saxe, Junteng Jia, Kalyan Vasuden Alwala, Karthik Prasad, Kartikeya Upasani, Kate Plawiak, Ke Li, Kenneth Heafield, Kevin Stone, Khalid El-Arini, Krithika Iyer, Kshitiz Malik, Kuenley Chiu, Kunal Bhalla, Kushal Lakhotia, Lauren Rantala-Yearly, Laurens van der Maaten, Lawrence Chen, Liang Tan, Liz Jenkins, Louis Martin, Lovish Madaan, Lubo Malo, Lukas Blecher, Lukas Landzaat, Luke de Oliveira, Madeline Muzzi, Mahesh Pasupuleti, Mannat Singh, Manohar Paluri, Marcin Kardas, Maria Tsimpoukelli, Mathew Oldham, Mathieu Rita, Maya Pavlova, Melanie Kambadur, Mike Lewis, Min Si, Mitesh Kumar Singh, Mona Hassan, Naman Goyal, Narjes Torabi, Nikolay Bashlykov, Nikolay Bogoychev, Niladri Chatterji, Ning Zhang, Olivier Duchenne, Onur Çelebi, Patrick Alrassy, Pengchuan Zhang, Pengwei Li, Petar Vasic, Peter Weng, Prajjwal Bhargava, Pratik Dubal, Praveen Krishnan, Punit Singh Koura, Puxin Xu, Qing He, Qingxiao Dong, Ragavan Srinivasan, Raj Ganapathy, Ramon Calderer, Ricardo Silveira Cabral, Robert Stojnic, Roberta Raileanu, Rohan Maheswari, Rohit Girdhar, Rohit Patel, Romain Sauvestre, Ronnie Polidoro, Roshan Sumbaly, Ross Taylor, Ruan Silva, Rui Hou, Rui Wang, Saghar Hosseini, Sahana Chennabasappa, Sanjay Singh, Sean Bell, Seohyun Sonia Kim, Sergey Edunov, Shaoliang Nie, Sharan Narang, Sharath Rapparthi, Sheng Shen, Shengye Wan, Shruti Bhosale, Shun Zhang, Simon Vandenhende, Soumya Batra, Spencer Whitman, Sten Sootla, Stéphane Collot, Suchin Gururangan, Sydney Borodinsky, Tamar Herman, Tara Fowler, Tarek Sheasha, Thomas Georgiou, Thomas Scialom, Tobias Speckbacher, Todor Mihaylov, Tong Xiao, Ujjwal Karn, Vedanuj Goswami, Vibhor Gupta, Vignesh Ramanathan, Viktor Kerkez, Vincent Gouget, Virginie Do, Vish Vogeti, Vitor Albiero, Vladan Petrovic, Weiwei Chu, Wenhan Xiong, Wenyan Fu, Whitney Meers, Xavier Martinet, Xiaodong Wang, Xiaofang Wang, Xiaoqing Ellen Tan, Xide Xia, Xinfeng Xie, Xuchao Jia, Xuwei Wang, Yaelle Goldschlag, Yashesh Gaur, Yasmine Babaei, Yi Wen, Yiwon Song, Yuchen Zhang, Yue Li, Yuning Mao, Zacharie Delpierre Coudert, Zheng Yan, Zhengxing Chen, Zoe Papakipos, Aaditya Singh, Aayushi Srivastava, Abha Jain, Adam Kelsey, Adam Shajnfeld, Adithya Gangidi, Adolfo Victoria, Ahuva Goldstand, Ajay Menon, Ajay Sharma, Alex Boesenberg, Alexei Baevski, Allie Feinstein, Amanda Kallet, Amit Sangani, Amos Teo, Anam Yunus, Andrei Lupu, Andres Alvarado, Andrew Caples, Andrew Gu, Andrew Ho, Andrew Poulton, Andrew Ryan, Ankit Ramchandani, Annie Dong, Annie Franco, Anuj Goyal, Aparajita Saraf, Arkabandhu Chowdhury, Ashley Gabriel, Ashwin Bharambe, Assaf Eisenman, Azadeh Yazdan, Beau James, Ben Maurer, Benjamin Leonhardi, Bernie Huang, Beth Loyd, Beto De Paola, Bhargavi Paran-

- jape, Bing Liu, Bo Wu, Boyu Ni, Braden Hancock, Bram Wasti, Brandon Spence, Brani Stojkovic, Brian Gamido, Britt Montalvo, Carl Parker, Carly Burton, Catalina Mejia, Ce Liu, Changhan Wang, Changkyu Kim, Chao Zhou, Chester Hu, Ching-Hsiang Chu, Chris Cai, Chris Tindal, Christoph Feichtenhofer, Cynthia Gao, Damon Civin, Dana Beaty, Daniel Kreymer, Daniel Li, David Adkins, David Xu, Davide Testuggine, Delia David, Devi Parikh, Diana Liskovich, Didem Foss, Dingkan Wang, Duc Le, Dustin Holland, Edward Dowling, Eissa Jamil, Elaine Montgomery, Eleonora Presani, Emily Hahn, Emily Wood, Eric-Tuan Le, Erik Brinkman, Esteban Arcaute, Evan Dunbar, Evan Smothers, Fei Sun, Felix Kreuk, Feng Tian, Filippas Kokkinos, Firat Ozgenel, Francesco Caggioni, Frank Kanayet, Frank Seide, Gabriela Medina Florez, Gabriella Schwarz, Gada Badeer, Georgia Sweet, Gil Halpern, Grant Herman, Grigory Sizov, Guangyi, Zhang, Guna Lakshminarayanan, Hakan Inan, Hamid Shojanazeri, Han Zou, Hannah Wang, Hanwen Zha, Haroun Habeeb, Harrison Rudolph, Helen Suk, Henry Aspegren, Hunter Goldman, Hongyuan Zhan, Ibrahim Damla, Igor Molybog, Igor Tufanov, Ilias Leontiadis, Irina-Elena Veliche, Itai Gat, Jake Weissman, James Geboski, James Kohli, Janice Lam, Japhet Asher, Jean-Baptiste Gaya, Jeff Marcus, Jeff Tang, Jennifer Chan, Jenny Zhen, Jeremy Reizenstein, Jeremy Teboul, Jessica Zhong, Jian Jin, Jingyi Yang, Joe Cummings, Jon Carvill, Jon Shepard, Jonathan McPhie, Jonathan Torres, Josh Ginsburg, Junjie Wang, Kai Wu, Kam Hou U, Karan Saxena, Kartikay Khandelwal, Katayoun Zand, Kathy Matosich, Kaushik Veeraraghavan, Kelly Michelen, Keqian Li, Kiran Jagadeesh, Kun Huang, Kunal Chawla, Kyle Huang, Lailin Chen, Lakshya Garg, Lavender A, Leandro Silva, Lee Bell, Lei Zhang, Liangpeng Guo, Licheng Yu, Liron Moshkovich, Luca Wehrstedt, Madian Khabza, Manav Avalani, Manish Bhatt, Martynas Mankus, Matan Hasson, Matthew Lennie, Matthias Reso, Maxim Groshev, Maxim Naumov, Maya Lathi, Meghan Keneally, Miao Liu, Michael L. Seltzer, Michal Valko, Michelle Restrepo, Mihir Patel, Mik Vyatskov, Mikayel Samvelyan, Mike Clark, Mike Macey, Mike Wang, Miquel Jubert Hermoso, Mo Metanat, Mohammad Rastegari, Munish Bansal, Nandhini Santhanam, Natascha Parks, Natasha White, Navyata Bawa, Nayan Singhal, Nick Egebo, Nicolas Usunier, Nikhil Mehta, Nikolay Pavlovich Laptev, Ning Dong, Norman Cheng, Oleg Chernoguz, Olivia Hart, Omkar Salpekar, Ozlem Kalinli, Parkin Kent, Parth Parekh, Paul Saab, Pavan Balaji, Pedro Rittner, Philip Bontrager, Pierre Roux, Piotr Dollar, Polina Zvyagina, Prashant Ratanachandani, Pritish Yuvraj, Qian Liang, Rachad Alao, Rachel Rodriguez, Rafi Ayub, Raghotham Murthy, Raghu Nayani, Rahul Mitra, Rangaprabhu Parthasarathy, Raymond Li, Rebekkah Hogan, Robin Battey, Rocky Wang, Russ Howes, Ruty Rinott, Sachin Mehta, Sachin Siby, Sai Jayesh Bondu, Samyak Datta, Sara Chugh, Sara Hunt, Sargun Dhillon, Sasha Sidorov, Satadru Pan, Saurabh Mahajan, Saurabh Verma, Seiji Yamamoto, Sharadh Ramaswamy, Shaun Lindsay, Shaun Mihailescu, Sheng Feng, Shenghao Lin, Shengxin Cindy Zha, Shishir Patil, Shiva Shankar, Shuqiang Zhang, Shuqiang Zhang, Sinong Wang, Sneha Agarwal, Soji Sajuyigbe, Soumith Chintala, Stephanie Max, Stephen Chen, Steve Kehoe, Steve Satterfield, Sudarshan Govindaprasad, Sumit Gupta, Summer Deng, Sungmin Cho, Sunny Virk, Suraj Subramanian, Sy Choudhury, Sydney Goldman, Tal Remez, Tamar Glaser, Tamara Best, Thilo Koehler, Thomas Robinson, Tianhe Li, Tianjun Zhang, Tim Matthews, Timothy Chou, Tzook Shaked, Varun Vontimitta, Victoria Ajayi, Victoria Montanez, Vijai Mohan, Vinay Satish Kumar, Vishal Mangla, Vlad Ionescu, Vlad Poenaru, Vlad Tiberiu Mihailescu, Vladimir Ivanov, Wei Li, Wenchen Wang, Wenwen Jiang, Wes Bouaziz, Will Constable, Xiaocheng Tang, Xiaojian Wu, Xiaolan Wang, Xilun Wu, Xinbo Gao, Yaniv Kleinman, Yanjun Chen, Ye Hu, Ye Jia, Ye Qi, Yenda Li, Yilin Zhang, Ying Zhang, Yossi Adi, Youngjin Nam, Yu, Wang, Yu Zhao, Yuchen Hao, Yundi Qian, Yunlu Li, Yuzi He, Zach Rait, Zachary DeVito, Zef Rosnbrick, Zhaoduo Wen, Zhenyu Yang, Zhiwei Zhao, and Zhiyu Ma. The Llama 3 Herd of Models, November 2024.
- [5] Zhenxiao Fu, Fan Chen, Shan Zhou, Haitong Li, and Lei Jiang. LLMCO2: Advancing accurate carbon footprint prediction for LLM inferences. *SIGENERGY Energy Inform. Rev.*, Vol. 5, No. 2, pp. 63–68, August 2025.
- [6] OpenAI, Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, Alex Iftimie, Alex Karpenko, Alex Tachard Passos, Alexander Neitz, Alexander Prokofiev, Alexander Wei, Allison Tam, Ally Bennett, Ananya Kumar, Andre Saraiva, Andrea Vallone, Andrew Duberstein, Andrew Kondrich, Andrey Mishchenko, Andy Applebaum, Angela Jiang, Ashvin Nair, Barret Zoph, Behrooz Ghorbani, Bohan Zhang, Ben Rossen, Benjamin Sokolowsky, Boaz Barak, Bob McGrew, Borys Minaiev, Botao Hao, Bowen Baker, Brandon Houghton, Brandon McKinzie, Brydon Eastman, Camillo Lugaresi, Cary Bassin, Cary Hudson, Chak Ming Li, Charles de Bourcy, Chelsea Voss, Chen Shen, Chong Zhang, Chris Koch, Chris Orsinger, Christopher Hesse, Claudia Fischer, Clive Chan, Dan Roberts, Daniel Kappler, Daniel Levy, Daniel Selsam, David Dohan, David Farhi, David Mely, David Robinson, Dimitris Tsipras, Doug Li, Dragos Oprica, Eben Freeman, Eddie Zhang, Edmund Wong, Elizabeth Proehl, Enoch Cheung, Eric Mitchell, Eric Wallace, Erik Ritter, Evan Mays, Fan Wang, Felipe Petroski Such, Filippo Raso, Florencia Leoni, Foivos Tsimpourlas, Francis Song, Fred von Lohmann, Freddie Sulit, Geoff Salmon, Giambattista Parascandolo, Gildas Chabot, Grace Zhao, Greg Brockman, Guillaume Leclerc, Hadi Salman, Haiming Bao, Hao Sheng, Hart Andrin, Hessam Bagherinezhad, Hongyu Ren, Hunter Lightman, Hyung Won Chung, Ian Kivlichan, Ian O’Connell,

- Ian Osband, Ignasi Clavera Gilaberte, Ilge Akkaya, Ilya Kostrikov, Ilya Sutskever, Irina Kofman, Jakub Pachocki, James Lennon, Jason Wei, Jean Harb, Jerry Twore, Jiacheng Feng, Jiahui Yu, Jiayi Weng, Jie Tang, Jieqi Yu, Joaquin Quiñero Candela, Joe Palermo, Joel Parish, Johannes Heidecke, John Hallman, John Rizzo, Jonathan Gordon, Jonathan Uesato, Jonathan Ward, Joost Huizinga, Julie Wang, Kai Chen, Kai Xiao, Karan Singhal, Karina Nguyen, Karl Cobbe, Katy Shi, Kayla Wood, Kendra Rimbach, Keren Gu-Lemberg, Kevin Liu, Kevin Lu, Kevin Stone, Kevin Yu, Lama Ahmad, Lauren Yang, Leo Liu, Leon Maksin, Leyton Ho, Liam Fedus, Lilian Weng, Linden Li, Lindsay McCallum, Lindsey Held, Lorenz Kuhn, Lukas Kondraciuk, Lukasz Kaiser, Luke Metz, Madelaine Boyd, Maja Trebacz, Manas Joglekar, Mark Chen, Marko Tintor, Mason Meyer, Matt Jones, Matt Kaufer, Max Schwarzer, Meghan Shah, Mehmet Yatbaz, Melody Y. Guan, Mengyuan Xu, Mengyuan Yan, Mia Glaese, Mianna Chen, Michael Lampe, Michael Malek, Michele Wang, Michelle Fradin, Mike McClay, Mikhail Pavlov, Miles Wang, Mingxuan Wang, Mira Murati, Mo Bavarian, Mostafa Rohaninejad, Nat McAleese, Neil Chowdhury, Neil Chowdhury, Nick Ryder, Nikolas Tezak, Noam Brown, Ofir Nachum, Oleg Boiko, Oleg Murk, Olivia Watkins, Patrick Chao, Paul Ashbourne, Pavel Izmailov, Peter Zhokhov, Rachel Dias, Rahul Arora, Randall Lin, Rapha Gontijo Lopes, Raz Gaon, Reah Miyara, Reimar Leike, Renny Hwang, Rhythm Garg, Robin Brown, Roshan James, Rui Shu, Ryan Cheu, Ryan Greene, Saachi Jain, Sam Altman, Sam Toizer, Sam Toyer, Samuel Miserendino, Sandhini Agarwal, Santiago Hernandez, Sasha Baker, Scott McKinney, Scottie Yan, Shengjia Zhao, Shengli Hu, Shibani Santurkar, Shraman Ray Chaudhuri, Shuyuan Zhang, Siyuan Fu, Spencer Papay, Steph Lin, Suchir Balaji, Suvansh Sanjeev, Szymon Sidor, Tal Broda, Aidan Clark, Tao Wang, Taylor Gordon, Ted Sanders, Tejal Patwardhan, Thibault Sottiaux, Thomas Degry, Thomas Dimson, Tianhao Zheng, Timur Garipov, Tom Stasi, Trapit Bansal, Trevor Creech, Troy Peterson, Tyna Eloundou, Valerie Qi, Vineet Kosaraju, Vinnie Monaco, Vitchyr Pong, Vlad Fomenko, Weiye Zheng, Wenda Zhou, Wenting Zhan, Wes McCabe, Wojciech Zaremba, Yann Dubois, Yinghai Lu, Yining Chen, Young Cha, Yu Bai, Yuchen He, Yuchen Zhang, Yunyun Wang, Zheng Shao, and Zhuohan Li. OpenAI o1 System Card, April 2026.
- [7] Angelo Jake. Uber burned through its entire 2026 AI budget in four months. Now its COO is questioning whether it's worth it.
 - [8] Qiang Cheng, Pengkai Lin, and Yue Zhao. Does generative AI facilitate investor Trading? Early evidence from ChatGPT outages. *Journal of Accounting and Economics*, Vol. 80, No. 2-3, p. 101821, November 2025.
 - [9] Stuart Geman, Elie Bienenstock, and René Doursat. Neural networks and the bias/variance dilemma. *Neural Computation*, Vol. 4, No. 1, pp. 1–58, 1992.
 - [10] Brady Neal, Sarthak Mittal, Aristide Baratin, Vinayak Tantia, Matthew Scicluna, Simon Lacoste-Julien, and Ioannis Mitliagkas. A Modern Take on the Bias-Variance Tradeoff in Neural Networks, December 2019.
 - [11] Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal. Reconciling modern machine learning practice and the bias-variance trade-off. *Proceedings of the National Academy of Sciences*, Vol. 116, No. 32, pp. 15849–15854, August 2019.
 - [12] Trevor Hastie, Robert Tibshirani, and J. H. Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, second edition edition, 2009.
 - [13] Danny Hernandez, Tom Brown, Tom Conerly, Nova DasSarma, Dawn Drain, Sheer El-Showk, Nelson Elhage, Zac Hatfield-Dodds, Tom Henighan, Tristan Hume, Scott Johnston, Ben Mann, Chris Olah, Catherine Olsson, Dario Amodei, Nicholas Joseph, Jared Kaplan, and Sam McCandlish. Scaling Laws and Interpretability of Learning from Repeated Data, May 2022.
 - [14] Aaron Tan. As AI costs spiral, Dell pitches return to on-premise datacentres.
 - [15] Moreno Javier. Augmented commerce: Machine learning at Shopify.
 - [16] Ziheng Wang, Jeremy Wohlwend, and Tao Lei. Structured Pruning of Large Language Models. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 6151–6162, Online, 2020. Association for Computational Linguistics.
 - [17] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network, March 2015.
 - [18] Junru Wu, Yue Wang, Zhenyu Wu, Zhangyang Wang, Ashok Veeraraghavan, and Yingyan Lin. Deep \$k\$-Means: Re-Training and Parameter Sharing with Harder Cluster Assignments for Compressing Deep Convolutions, June 2018.
 - [19] Benoit Jacob, Skirmantas Kligys, Bo Chen, Menglong Zhu, Matthew Tang, Andrew Howard, Hartwig Adam, and Dmitry Kalenichenko. Quantization and Training of Neural Networks for Efficient Integer-Arithmetic-Only Inference, December 2017.
 - [20] Markus Nagel, Rana Ali Amjad, Mart van Baalen, Christos Louizos, and Tijmen Blankevoort. Up or Down? Adaptive Rounding for Post-Training Quantization, June 2020.
 - [21] Jeremy M. Cohen, Alex Damian, Ameeta Talwalkar, J. Zico Kolter, and Jason D. Lee. Understanding Optimization in Deep Learning with Central Flows, September 2025.
 - [22] C. E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, Vol. 27, No. 3, pp. 379–423, 1948.

- [23] B. Ebanks, P.K. Sahoo, and W. Sander. *Characterizations of Information Measures*. World Scientific Publishing Company, 1998.
- [24] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 Technical Report, May 2025.
- [25] Qwen Team. Qwen3.5: Accelerating productivity with native multimodal agents, February 2026.
- [26] Gemma Team, Sherif El Abd, Vaibhav Aggarwal, Robin Algayres, Alek Andreev, Olivier Bachem, Ian Ballantyne, Cormac Brick, Victor Cărbune, Michelle Casbon, Mayank Chaturvedi, Aditya Chawla, Victor Cotruta, Alice Coucke, Phil Culliton, Robert Dadashi, Lucas Dixon, Mohamed Elhawaty, Utku Evci, Clément Farabet, Johan Ferret, Filippo Galgani, Sertan Girgin, Jean-Bastien Grill, Maarten Grootendorst, Jiaxian Guo, Cassidy Hardin, Yanzhang He, Steven M. Hernandez, Omri Homburger, Léonard Hussenot, Juyeong Ji, Armand Joulin, Aishwarya Kamath, Parnian Kassarai, Olivier Lacombe, Preethi Lahoti, Gaël Liu, Gus Martins, Luciano Martins, Tatiana Matejovicova, Ramona Merhej, Nikola Momchev, Sneha Mondal, Ryan Mullins, Sindhu Raghuram Panyam, Shreya Pathak, Sarah Perrin, André Susano Pinto, Etienne Pot, Angéline Pouget, Alexandre Ramé, Sabela Ramos, Douglas Reid, David Rim, Morgane Rivière, Karsten Roth, Louis Rouillard, Omar Sanseviero, Pier Giuseppe Sessa, Shane Settle, Danila Sinopalnikov, Sara Smoot, Piotr Stanczyk, Andreas Steiner, Lawrence Stewart, Ilya Tolstikhin, Michael Tschannen, Anton Tsitsulin, Nino Vieillard, Renjie Wu, Pingmei Xu, Haichuan Yang, Edouard Yvinec, Biao Zhang, Li Zhang, Joe Zou, Nicolas Aagnes, Abdelrahman Abdelhamed, Jakub Adamek, Shivani Agrawal, Shubham Agrawal, Ibrahim Alabdulmohsin, Jean Baptiste Alayrac, Uri Alon, Chandramouli Amarnath, Ankesh Anand, Chrysovalantis Anastasiou, Setareh Ariafar, François-Xavier Aubet, Kyriakos Axiotis, Federico Barbero, Joelle Barral, Alexei Bendebury, Urs Bergmann, Stanley Bileschi, Kat Black, Mathieu Blondel, Sebastian Borgeaud, Arthur Brazinskas, Ryan Burnell, Robert Busa-Fekete, Mu Cai, Daniele Calandriello, Glenn Cameron, Charlotte Caucheteux, Rahma Chaabouni, Garima Chadha, Jetha Chan, Blake Jianhang Chen, Jesse Chen, Lin Chen, Xu Chen, Derek Cheng, Tzu-hsiang Chien, Nikolai Chinaev, Yi Chou, Zhaoxue Chu, Benjamin Coleman, Pooja Consul, Sam Conway-Rahman, Scott Crowell, Dylan Cutler, Vivek Dani, Samira Daruki, Anil Das, Daniel Deutsch, Nishanth Dikkala, Li Ding, Qiuhan Ding, Shenil Dodhia, Konstantin Donhauser, Tulsee Doshi, Anca Dragan, Alex Druinsky, Sahil Dua, Zoltan Egyed, Danielle Eisenbud, Daniel Eppens, Cindy Fan, Bahare Fatemi, Yassir Fathullah, Vlad Feinberg, Milen Ferev, Sebastian Flennerhag, Takumi Fujimoto, João Gabriel Oliveira, Isaac Galatzer-Levy, João Gante, Simon Geisler, Soham Ghosal, Antonios M. Girgis, Tamara von Glehn, Alec Go, Alhaad Gokhale, Alex Grills, Yiming Gu, Mayank Gupta, Pramod Gupta, Guru Guruganesh, Raia Hadsell, Hamza Harkous, Jitendra Harlalka, Demis Hassabis, Anja Hauth, Joe Heyward, Arian Hosseini, Chih-Yang Hsia, I-Hung Hsu, Xiaopeng Huang, Yangsibo Huang, Kevin Hui, Adrian Hutter, Te I, Fotis Iliopoulos, Advait Jain, Ganesh Jawahar, Ziwei Ji, Qilin Jin, Melvin Johnson, Kandarp Joshi, Arun Kandoor, Wang-Cheng Kang, Koray Kavukcuoglu, Mehran Kazemi, Kathleen Kenealy, Amr Khalifa, Phoebe Kirk, Ivan Korotkov, Suraj Kothawade, Vitaly Kovalev, Neel Kovelamudi, Adam Kraft, Ravin Kumar, Vivek Kumar, Harish Kuppam, Justin Lannin, Chen-Yu Lee, Seungji Lee, Dmitry Lepikhin, Alon Levkovitch, Dongdong Li, Qiuqia Li, Valentin Liévin, Ethan Lin, Ziqian Lin, Casper Liu, Tianlin Liu, Tianqi Liu, Xin Liu, Ivan Lobov, Mayank Lunayach, Min Ma, Gagan Madan, Andrii Maksai, Eric Malmi, Michal Matuszak, Daniel McDuff, Gaurav Menghani, Maciej Mikula, Daniil Mirylenka, Karolis Misiunas, Vedant Misra, Andreea Mitran, Kareem Mohamed, Maksim Mukha, Eric Noland, James O'Donnell, Brendan O'Donoghue, Kate Olszewska, Bennett Orlando, Wanqiong Pan, Rina Panigrahy, Unnati Parekh, Nicolas Perez-Nieves, Chunjong Park, Eric Paskie, Liqian Peng, Bryce Pettrini, Slav Petrov, Jonas Pfeiffer, Bilal Piot, Martyna Plomecka, Siim Poder, Octavio Ponce, Arijit Pramanik, David Racz, Anish Rajan, Michelle Ramanovich, Anand Rao, Marvin Ritter, Vitor Rodrigues, Evan Rosen, Mikołaj Rybiński, Naveen Sachdeva, Michaël E. Sander, Rohit Sathyanarayana, Sagar Savla, Samuel Schmidgall, Tal Schuster, George Scrivener, Benoit Seguin, Andrew Sellergren, Aliaksei Severyn, Izhak Shafran, Dhruv Shah, Bobak Shahriari, Yuan Shangguan, Ashish Shenoy, Pradeep Shenoy, Rakesh Shivanna, Pauline Sho, Lucas Spangher, Wojciech Stokowiec, Tim Strother, Yao Su, Yinghao Sun, Mukund Sundararajan, Andrea Tacchetti, Mor Hazan Taege, Pouya Tafti, Jean Tarbouriech, Chetan Tekur, Shantanu Thakoor, Rahul Thapa, Madeleine Traverse, Lenart Treven, Tao Tu, Chien Te Tung, Çağlar Ünlü, Petar Velicković, Malini Pooni Venkat, Sagar Gubbi Venkatesh, Vidya Venkiteswaran, Francesco Visin, Alex Vitvitskyi, Kiran Vodrahalli, Weiye Wang, Xin Wang, Tris Warkentin, Jan Wassenberg, John Wieting, Cindy Wu, Lechao Xiao, Hao Xu, Yuhui Xu, Fuzhao Xue, Arun Yadav, Jun Yan, Antoine Yang, Lin Yang, Ming-Hsuan Yang, Ziyu Ying, Jae Hyeon Yoo, Morteza Zadi-

- moghaddam, Sajjad Zafar, Fred Zhang, Jiageng Zhang, Jianyi Zhang, Xiaofan Zhang, Chao Zhao, David Zhou, and Chen Zou. Gemma 4 Technical Report, July 2026.
- [27] Z.ai inc. GLM-4.7: Advancing the Coding Capability, December 2025.
- [28] Samuel L. Smith, Benoit Dherin, David G. T. Barrett, and Soham De. On the Origin of Implicit Regularization in Stochastic Gradient Descent, 2021.
- [29] Ronen Eldan and Yuanzhi Li. TinyStories: How Small Can Language Models Be and Still Speak Coherent English?, May 2023.
- [30] Sid Black, Leo Gao, Phil Wang, Connor Leahy, and Stella Biderman. GPT-Neo: Large scale autoregressive language modeling with meshtensorflow. Zenodo, October 2021.
- [31] Alec Radford, Jeff Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. 2019.
- [32] Taku Kudo and John Richardson. SentencePiece: A simple and language independent subword tokenizer and detokenizer for Neural Text Processing, August 2018.
- [33] Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and Efficient Foundation Language Models, February 2023.
- [34] F. Jelinek, R. L. Mercer, L. R. Bahl, and J. K. Baker. Perplexity—a measure of the difficulty of speech recognition tasks. *The Journal of the Acoustical Society of America*, Vol. 62, No. S1, pp. S63–S63, December 1977.
- [35] Markus Nagel, Marios Fournarakis, Rana Ali Amjad, Yelysei Bondarenko, Mart van Baalen, and Tijmen Blankevoort. A White Paper on Neural Network Quantization, June 2021.
- [36] S. Kullback and R. A. Leibler. On Information and Sufficiency. *The Annals of Mathematical Statistics*, Vol. 22, No. 1, pp. 79–86, March 1951.
- [37] Augustin Cauchy. Méthode générale pour la résolution des systèmes d’équations simultanées. *Comp. Rend. Sci. Paris*, Vol. 25, pp. 536–538, 1847.
- [38] Jeremy M. Cohen, Simran Kaur, Yuanzhi Li, J. Zico Kolter, and Ameet Talwalkar. Gradient Descent on Neural Networks Typically Occurs at the Edge of Stability, November 2022.
- [39] Kimi Team, Yifan Bai, Yiping Bao, Y. Charles, Cheng Chen, Guanduo Chen, Haiting Chen, Huarong Chen, Jiahao Chen, Ningxin Chen, Ruijie Chen, Yanru Chen, Yuankun Chen, Yutian Chen, Zhuofu Chen, Jialei Cui, Hao Ding, Mengnan Dong, Angang Du, Chenzhuang Du, Dikang Du, Yulun Du, Yu Fan, Yichen Feng, Kelin Fu, Bofei Gao, Chenxiao Gao, Hongcheng Gao, Peizhong Gao, Tong Gao, Yuyao Ge, Shangyi Geng, Qizheng Gu, Xinran Gu, Longyu Guan, Haiqing Guo, Jianhang Guo, Xiaoru Hao, Tianhong He, Weiran He, Wenyang He, Yunjia He, Chao Hong, Hao Hu, Yangyang Hu, Zhenxing Hu, Weixiao Huang, Zhiqi Huang, Zihao Huang, Tao Jiang, Zhejun Jiang, Xinyi Jin, Yongsheng Kang, Guokun Lai, Cheng Li, Fang Li, Haoyang Li, Ming Li, Wentao Li, Yang Li, Yanhao Li, Yiwei Li, Zhaowei Li, Zheming Li, Hongzhan Lin, Xiaohan Lin, Zongyu Lin, Chengyin Liu, Chenyu Liu, Hongzhang Liu, Jingyuan Liu, Junqi Liu, Liang Liu, Shaowei Liu, T. Y. Liu, Tianwei Liu, Weizhou Liu, Yangyang Liu, Yibo Liu, Yiping Liu, Yue Liu, Zhengying Liu, Enzhe Lu, Haoyu Lu, Lijun Lu, Yashuo Luo, Shengling Ma, Xinyu Ma, Yingwei Ma, Shaoguang Mao, Jie Mei, Xin Men, Yibo Miao, Siyuan Pan, Yebo Peng, Ruoyu Qin, Zeyu Qin, Bowen Qu, Zeyu Shang, Lidong Shi, Shengyuan Shi, Feifan Song, Jianlin Su, Zhengyuan Su, Lin Sui, Xinjie Sun, Flood Sung, Yunpeng Tai, Heyi Tang, Jiawen Tao, Qifeng Teng, Chaoran Tian, Chensi Wang, Dinglu Wang, Feng Wang, Hailong Wang, Haiming Wang, Jianzhou Wang, Jiaxing Wang, Jinhong Wang, Shengjie Wang, Shuyi Wang, Si Wang, Xinyuan Wang, Yao Wang, Yejie Wang, Yiqin Wang, Yuxin Wang, Yuzhi Wang, Zhaoji Wang, Zhengtao Wang, Zhengtao Wang, Chu Wei, Qianqian Wei, Haoning Wu, Wenhao Wu, Xingzhe Wu, Yuxin Wu, Chenjun Xiao, Jin Xie, Xiaotong Xie, Weimin Xiong, Boyu Xu, Jinjing Xu, L. H. Xu, Lin Xu, Suting Xu, Weixin Xu, Xinran Xu, Yangchuan Xu, Ziyao Xu, Jing Xu, Jing Xu, Junjie Yan, Yuzi Yan, Hao Yang, Xiaofei Yang, Yi Yang, Ying Yang, Zhen Yang, Zhilin Yang, Zonghan Yang, Haotian Yao, Xingcheng Yao, Wenjie Ye, Zhuorui Ye, Bohong Yin, Longhui Yu, Enming Yuan, Hongbang Yuan, Mengjie Yuan, Siyu Yuan, Haobing Zhan, Dehao Zhang, Hao Zhang, Wanlu Zhang, Xiaobin Zhang, Yadong Zhang, Yangkun Zhang, Yichi Zhang, Yizhi Zhang, Yongting Zhang, Yu Zhang, Yutao Zhang, Yutong Zhang, Zheng Zhang, Haotian Zhao, Yikai Zhao, Zijia Zhao, Huabin Zheng, Shaojie Zheng, Longguang Zhong, Jianren Zhou, Xinyu Zhou, Zaida Zhou, Jinguo Zhu, Zhen Zhu, Weiyu Zhuang, and Xinxing Zu. Kimi K2: Open Agentic Intelligence, February 2026.
- [40] Chen Xing, Devansh Arpit, Christos Tsirigotis, and Yoshua Bengio. A Walk with SGD, May 2018.
- [41] Adam Paszke, Sam Gross, Francisco Massa, Adam Lerer, James Bradbury, Gregory Chanan, Trevor Killeen, Zeming Lin, Natalia Gimelshein, Luca Antiga, Alban Desmaison, Andreas Köpf, Edward Yang, Zach DeVito, Martin Raison, Alykhan Tejani, Sasank Chilamkurthy, Benoit Steiner, Lu Fang, Junjie Bai, and Soumith Chintala. PyTorch: An Imperative Style, High-Performance Deep Learning Library, December 2019.
- [42] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexan-

- dre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in python. *Journal of Machine Learning Research*, Vol. 12, No. 85, pp. 2825–2830, 2011.
- [43] Daniel S. Kermany, Michael Goldbaum, Wenjia Cai, Carolina C.S. Valentim, Huiying Liang, Sally L. Baxter, Alex McKeown, Ge Yang, Xiaokang Wu, Fangbing Yan, Justin Dong, Made K. Prasadha, Jacqueline Pei, Magdalene Y.L. Ting, Jie Zhu, Christina Li, Sierra Hewett, Jason Dong, Ian Ziyar, Alexander Shi, Runze Zhang, Lianghong Zheng, Rui Hou, William Shi, Xin Fu, Yaou Duan, Viet A.N. Huu, Cindy Wen, Edward D. Zhang, Charlotte L. Zhang, Oulan Li, Xiaobo Wang, Michael A. Singer, Xiaodong Sun, Jie Xu, Ali Tafreshi, M. Anthony Lewis, Huimin Xia, and Kang Zhang. Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning. *Cell*, Vol. 172, No. 5, pp. 1122–1131.e9, February 2018.
- [44] E. Borenstein, E. Sharon, and Shimon Ullman. Combining top-down and bottom-up segmentation. Vol. 2004, pp. 46–46, July 2004.
- [45] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization, January 2017.
- [46] Ilya Loshchilov and Frank Hutter. SGDR: Stochastic Gradient Descent with Warm Restarts, May 2017.
- [47] Leonid I. Rudin, Stanley Osher, and Emad Fatemi. Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, Vol. 60, No. 1-4, pp. 259–268, November 1992.
- [48] Dmitry Ulyanov, Andrea Vedaldi, and Victor Lempitsky. Deep Image Prior. *International Journal of Computer Vision*, Vol. 128, No. 7, pp. 1867–1888, July 2020.
- [49] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization, January 2019.